

Sliper — QML / C++ Kaynak Kodu

Proje: Sliper_QML_C++

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main.cpp

```
1 #include <QqmlContext>
2 #include <QApplication>
3 #include <QqmlApplicationEngine>
4 #include <QDebug>
5 #include <QtGlobal>
6 #include "src/Backend.h"
7 #include "src/SensorManager.h"
8 #include "src/Calculator.h"
9 #include "src/Database.h"
10 #include "src/ReportManager.h"
11 #include "src/WifiManager.h"
12
13 int main(int argc, char *argv[])
14 {
15     qputenv("QT_QUICK_CONTROLS_STYLE", "Basic");
16
17     QApplication app(argc, argv);
18     QApplication::setApplicationName("SLIPER");
19
20     Backend backend;
21     SensorManager sensorManager;
22     Calculator calculator;
23     Database database;
24     ReportManager reportManager;
25     WifiManager wifiManager(&sensorManager, &database);
26
27     QqmlApplicationEngine engine;
28
29     engine.rootContext()->setContextProperty("backend", &backend);
30     engine.rootContext()->setContextProperty("sensorManager", &sensorManager);
31     engine.rootContext()->setContextProperty("calculator", &calculator);
```

```
32     engine.rootContext()->setContextProperty("database", &database);
33     engine.rootContext()->setContextProperty("reportManager", &reportManager);
34     engine.rootContext()->setContextProperty("wifiManager", &wifiManager);
35
36     QObject::connect(
37         &engine,
38         &QQmlApplicationEngine::objectCreationFailed,
39         &app,
40         []() {
41             qWarning() << "QML nesnesi olusturulamadi!";
42             QCoreApplication::exit(-1);
43         },
44         Qt::QueuedConnection);
45
46     engine.loadFromModule("sliper", "Main");
47
48     return app.exec();
49 }
```

qml/Main.qml

```
1 import QtQuick 6.7
2 import QtQuick.Controls 6.7
3 import QtQuick.Window 6.7
4 import QtQuick.Layouts 6.7
5 import "pages"
6
7 ApplicationWindow {
8     id: pencere
9     visibility: Window.Maximized
10    title: "SLIPER Analiz"
11
12    background: Rectangle {
13        color: "#060d17"
14    }
15
16    StackLayout {
17        id: anaStack
18        anchors.fill: parent
19        currentIndex: 0
20
21        LoginPage {
22            onGirisBasarili: anaStack.currentIndex = 1
23        }
24
25        DashboardPage {
26        }
27    }
28 }
```

qml/components/SuTerazisi.qml

```
1 import QtQuick 6.7
2
3 Item {
4     id: root
5     property real egimX: 0
6     property real egimY: 0
7     property real maxAci: 15
8
9     width: 200
10    height: 200
11
12    readonly property real yaricap: (width / 2) - 20
13    readonly property bool duz: Math.abs(egimX) < 0.5 && Math.abs(egimY) < 0.5
14
15    readonly property real kabarcikOffsetX: Math.max(-1, Math.min(1, egimY / maxAci)) * yaricap
16    readonly property real kabarcikOffsetY: Math.max(-1, Math.min(1, egimX / maxAci)) * yaricap
17
18    Rectangle {
19        anchors.fill: parent
20        radius: width / 2
21        color: "#0f1420"
22        border.color: "#1e2a3f"
23        border.width: 2
24    }
25
26    Rectangle {
27        anchors.centerIn: parent
28        width: root.yaricap * 1.2
29        height: width
30        radius: width / 2
31        color: "transparent"
```

```
32     border.color: "#1e2a3f"
33     border.width: 1
34 }
35
36 Rectangle {
37     anchors.centerIn: parent
38     width: parent.width - 20
39     height: 1
40     color: "#1e2a3f"
41 }
42
43 Rectangle {
44     anchors.centerIn: parent
45     width: 1
46     height: parent.height - 20
47     color: "#1e2a3f"
48 }
49
50 Rectangle {
51     id: kabarcik
52     width: 28
53     height: 28
54     radius: 14
55     color: root.duz ? "#16a34a" : "#dc2626"
56     border.color: root.duz ? "#4ade80" : "#f87171"
57     border.width: 2
58
59     x: (root.width / 2) - (width / 2) + root.kabarcikOffsetX
60     y: (root.height / 2) - (height / 2) + root.kabarcikOffsetY
61
62     Behavior on x { NumberAnimation { duration: 150; easing.type: Easing.OutQuad } }
63     Behavior on y { NumberAnimation { duration: 150; easing.type: Easing.OutQuad } }
64     Behavior on color { ColorAnimation { duration: 200 } }
```

```
65         Behavior on border.color { ColorAnimation { duration: 200 } }
66     }
67
68     Text {
69         anchors.top: parent.bottom
70         anchors.horizontalCenter: parent.horizontalCenter
71         anchors.topMargin: 8
72         text: root.duz ? "Düz" : "Ayarlanıyor..."
73         color: root.duz ? "#4ade80" : "#6b7280"
74         font.family: "Segoe UI"
75         font.pixelSize: 12
76         font.bold: root.duz
77     }
78 }
```


qml/pages/DashboardPage.qml

```
1 import QtQuick 6.7
2 import QtQuick.Controls 6.7
3 import QtQuick.Layouts 6.7
4
5 Rectangle {
6     color: "#0a0e17"
7
8     readonly property var navOgeleri: [
9         { etiket: "Ölçüm", ikon: "📊" },
10        { etiket: "Sonuçlar", ikon: "📝" },
11        { etiket: "Geçmiş", ikon: "🕒" },
12        { etiket: "Kalibrasyon", ikon: "⚙️" }
13    ]
14
15    Row {
16        anchors.fill: parent
17        spacing: 0
18
19        Rectangle {
20            id: sidebar
21            width: 230
22            height: parent.height
23
24            gradient: Gradient {
25                GradientStop { position: 0.0; color: "#121b30" }
26                GradientStop { position: 1.0; color: "#080b12" }
27            }
28
29            Image {
30                anchors.top: parent.top
31                anchors.horizontalCenter: parent.horizontalCenter
```

```

32         anchors.topMargin: 24
33         source: "../assets/logo.png"
34         width: 144
35         height: 144
36         fillMode: Image.PreserveAspectFit
37     }
38
39     Rectangle {
40         id: baglantiDurumu
41         anchors.top: parent.top
42         anchors.left: parent.left
43         anchors.right: parent.right
44         anchors.leftMargin: 24
45         anchors.rightMargin: 24
46         anchors.topMargin: 178
47         height: 36
48         radius: 6
49         color: baglantiAlani.hoverli ? "#101a2c" : "transparent"
50         Behavior on color { ColorAnimation { duration: 120 } }
51
52     Row {
53         anchors.left: parent.left
54         anchors.leftMargin: 8
55         anchors.verticalCenter: parent.verticalCenter
56         spacing: 8
57
58         Rectangle {
59             width: 8
60             height: 8
61             radius: 4
62             anchors.verticalCenter: parent.verticalCenter
63             color: wifiManager.baglandi ? "#16a34a" : (wifiManager.baglaniyor ? "#f59e0b" :
"#dc2626")
64         }

```

```
65
66         Text {
67             text: wifiManager.baglandi ? "SLIPER-ESP32 Bağlı" : (wifiManager.baglaniyor ?
"Bağlanıyor..." : "Bağlı Değil")
68             color: "#9ca3af"
69             font.family: "Segoe UI"
70             font.pixelSize: 12
71             anchors.verticalCenter: parent.verticalCenter
72         }
73     }
74
75     MouseArea {
76         id: baglantiAlani
77         anchors.fill: parent
78         hoverEnabled: true
79         cursorShape: Qt.PointingHandCursor
80         enabled: !wifiManager.baglandi && !wifiManager.baglaniyor
81         onClicked: wifiManager.baglan()
82     }
83 }
84
85 Rectangle {
86     anchors.top: parent.top
87     anchors.left: parent.left
88     anchors.right: parent.right
89     anchors.leftMargin: 24
90     anchors.rightMargin: 24
91     anchors.topMargin: 224
92     height: 1
93     color: "#1f2c46"
94 }
95
96 Column {
```

```
97         anchors.top: parent.top
98
99         anchors.topMargin: 242
100
101         width: parent.width
102
103         spacing: 2
104
105     Repeater {
106
107         model: navOgeleri
108
109
110         Rectangle {
111
112             id: navOgesi
113
114             width: parent.width
115
116             height: 46
117
118             property bool aktif: icerikStack.currentIndex === index
119
120             property bool hoverli: false
121
122             color: aktif ? "#182644" : (hoverli ? "#101a2c" : "transparent")
123
124             Behavior on color { ColorAnimation { duration: 120 } }
125
126
127             Rectangle {
128
129                 width: 3
130
131                 height: parent.height
132
133                 color: "#3b82f6"
134
135                 visible: navOgesi.aktif
136
137             }
138
139
140         Row {
141
142             anchors.left: parent.left
143
144             anchors.leftMargin: 24
145
146             anchors.verticalCenter: parent.verticalCenter
147
148             spacing: 12
149
150
151             Text {
152
153                 text: modelData.ikon
154
155                 font.pixelSize: 15
```

```

130             anchors.verticalCenter: parent.verticalCenter
131         }
132
133         Text {
134             text: modelData.etiket
135             color: navOgesi.aktif ? "#6ea8ff" : (navOgesi.hoverli ? "#dce8f5" :
"#9ca3af")
136             font.family: "Segoe UI"
137             font.pixelSize: 14
138             font.bold: navOgesi.aktif
139             anchors.verticalCenter: parent.verticalCenter
140             Behavior on color { ColorAnimation { duration: 120 } }
141         }
142     }
143
144     MouseArea {
145         anchors.fill: parent
146         hoverEnabled: true
147         cursorShape: Qt.PointingHandCursor
148         onEntered: navOgesi.hoverli = true
149         onExited: navOgesi.hoverli = false
150         onClicked: icerikStack.currentIndex = index
151     }
152 }
153 }
154 }
155 }
156
157 Rectangle {
158     id: sidebarAyraci
159     width: 3
160     height: parent.height
161     color: "#05070b"
162

```

```
163         Rectangle {
164             anchors.right: parent.right
165             anchors.top: parent.top
166             anchors.bottom: parent.bottom
167             width: 1
168             color: "#403b82f6"
169         }
170     }
171
172     StackLayout {
173         id: icerikStack
174         width: parent.width - sidebar.width - sidebarAyraci.width
175         height: parent.height
176         currentIndex: 0
177
178         OlcumPage {
179             onOlcumTamamlandi: function(id) {
180                 sonuclarSayfasi.olcumId = id
181                 icerikStack.currentIndex = 1
182             }
183         }
184
185         SonuclarPage {
186             id: sonuclarSayfasi
187         }
188
189         GecmisPage {
190             onTestSecildi: function(id) {
191                 sonuclarSayfasi.olcumId = id
192                 icerikStack.currentIndex = 1
193             }
194         }
195     }
```

```
196         KalibrasyonPage {}
```

```
197     }
```

```
198 }
```

```
199 }
```

```
1 import QtQuick 6.7
2 import QtQuick.Controls 6.7
3
4 Rectangle {
5     id: gecmisSayfasi
6     color: "#0a0e17"
7
8     signal testSecildi(int id)
9
10    property var olcumListesi: []
11
12    function listeyiYenile() {
13        olcumListesi = database.tumOlcumleriGetir()
14    }
15
16    onVisibleChanged: {
17        if (visible) {
18            listeyiYenile()
19        }
20    }
21
22    Component.onCompleted: listeyiYenile()
23
24    Column {
25        anchors.fill: parent
26        anchors.margins: 20
27        spacing: 16
28
29        Row {
30            width: parent.width
31            spacing: 12
```



```
32
33     TextField {
34         id: aramaKutusu
35         width: parent.width - filtreKutusu.width - parent.spacing
36         height: 40
37         placeholderText: "Müşteri veya reçete ara..."
38         placeholderTextColor: "#4b5563"
39         color: "ffffff"
40         font.pixelSize: 13
41         leftPadding: 34
42         verticalAlignment: TextInput.AlignVCenter
43         background: Rectangle {
44             color: "#0f1420"
45             radius: 8
46             border.color: aramaKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
47             border.width: 1
48         }
49
50         Text {
51             anchors.left: parent.left
52             anchors.leftMargin: 12
53             anchors.verticalCenter: parent.verticalCenter
54             text: "🔍"
55             color: "#4b5563"
56             font.pixelSize: 13
57         }
58     }
59
60     ComboBox {
61         id: filtreKutusu
62         width: 160
63         height: 40
64         model: ["Tüm Testler", "Son 7 Gün", "Son 30 Gün"]
```

```
65
66         background: Rectangle {
67             color: "#0f1420"
68             radius: 8
69             border.color: filtreKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
70             border.width: 1
71         }
72
73         contentItem: Text {
74             text: filtreKutusu.displayText
75             color: "#9ca3af"
76             font.pixelSize: 13
77             leftPadding: 10
78             verticalAlignment: Text.AlignVCenter
79         }
80     }
81 }
82
83 ListView {
84     width: parent.width
85     height: parent.height - 56
86     clip: true
87     spacing: 10
88
89     model: gecmisSayfasi.olcumListesi
90
91     delegate: Rectangle {
92         id: gecmisKarti
93         width: parent.width
94         height: 76
95         radius: 10
96         color: "#0f1420"
97         border.color: kartAlani.containsMouse ? "#3b82f6" : "#1e2a3f"
```

```
98         border.width: 1
99         Behavior on border.color { ColorAnimation { duration: 120 } }
100
101         MouseArea {
102             id: kartAlani
103             anchors.fill: parent
104             hoverEnabled: true
105             acceptedButtons: Qt.NoButton
106         }
107
108         Rectangle {
109             id: rozet
110             width: 44
111             height: 44
112             radius: 10
113             anchors.left: parent.left
114             anchors.verticalCenter: parent.verticalCenter
115             anchors.leftMargin: 16
116             color: "#132335"
117
118             Text {
119                 anchors.centerIn: parent
120                 text: modelData.musteri.charAt(0).toUpperCase()
121                 color: "#3b82f6"
122                 font.family: "Segoe UI"
123                 font.pixelSize: 18
124                 font.bold: true
125             }
126         }
127
128         Column {
129             anchors.left: rozet.right
130             anchors.verticalCenter: parent.verticalCenter
```

```
131         anchors.leftMargin: 14
132         spacing: 4
133
134         Row {
135             spacing: 8
136
137             Text {
138                 text: modelData.tarih
139                 color: "#6b7280"
140                 font.family: "Segoe UI"
141                 font.pixelSize: 11
142             }
143
144             Rectangle {
145                 width: strokeMetni.implicitWidth + 12
146                 height: 16
147                 radius: 8
148                 color: "#2a2010"
149                 anchors.verticalCenter: parent.verticalCenter
150
151                 Text {
152                     id: strokeMetni
153                     anchors.centerIn: parent
154                     text: modelData.strokeSayisi + " stroke"
155                     color: "#e8a020"
156                     font.family: "Segoe UI"
157                     font.pixelSize: 10
158                     font.bold: true
159                 }
160             }
161         }
162
163         Text {
```

```
164         text: "Müşteri: " + modelData.musteri + " • Reçete: " + modelData.recete
165         color: "#dce8f5"
166         font.family: "Segoe UI"
167         font.pixelSize: 13
168         font.bold: true
169     }
170 }
171
172 Button {
173     anchors.right: parent.right
174     anchors.verticalCenter: parent.verticalCenter
175     anchors.rightMargin: 16
176     height: 34
177     text: "Görüntüle"
178     font.pixelSize: 12
179
180     onClicked: testSecildi(modelData.id)
181
182     background: Rectangle {
183         radius: 6
184         color: parent.hovered ? "#4f8cf7" : "#3b82f6"
185         Behavior on color { ColorAnimation { duration: 120 } }
186     }
187
188     contentItem: Text {
189         text: parent.text
190         color: "#ffffff"
191         font: parent.font
192         horizontalAlignment: Text.AlignHCenter
193         verticalAlignment: Text.AlignVCenter
194         leftPadding: 8
195         rightPadding: 8
196     }
```

197 }

198 }

199 }

200 }

201 }

```
1 import QtQuick 6.7
2 import QtQuick.Controls 6.7
3 import QtQuick.Layouts 6.7
4 import "../components"
5
6 Rectangle {
7     color: "#0a0e17"
8
9     property int seciliSensor: -1
10    property int egimAdimi: 0
11    property int loadCellAdimi: 0
12    property int mesafeAdimi: 0
13
14    property bool egimKayitliVar: false
15    property string egimSonTarih: ""
16
17    property bool loadCellKayitliVar: false
18    property real loadCellReferans: 0
19    property string loadCellSonTarih: ""
20    property real loadCellSifirHam: 0
21
22    property bool mesafeUstKayitliVar: false
23    property bool mesafeAltKayitliVar: false
24    property string mesafeSonTarih: ""
25
26    onSeciliSensorChanged: {
27        if (seciliSensor === 0) {
28            var e = database.kalibrasyonGetir("egim")
29            egimKayitliVar = e.mevcut
30            egimSonTarih = e.mevcut ? e.tarih : ""
31        } else if (seciliSensor === 1) {
```

```
32         var l = database.kalibrasyonGetir("loadcell")
33         loadCellKayitliVar = l.mevcut
34         loadCellReferans = l.mevcut ? l.deger1 : 0
35         loadCellSonTarih = l.mevcut ? l.tarih : ""
36     } else if (seciliSensor === 2) {
37         var u = database.kalibrasyonGetir("mesafe_ust")
38         var a = database.kalibrasyonGetir("mesafe_alt")
39         mesafeUstKayitliVar = u.mevcut
40         mesafeAltKayitliVar = a.mevcut
41         mesafeSonTarih = u.mevcut ? u.tarih : (a.mevcut ? a.tarih : "")
42     }
43 }
44
45 StackLayout {
46     anchors.fill: parent
47     currentIndex: seciliSensor === -1 ? 0 : 1
48
49     Item {
50         Text {
51             anchors.top: parent.top
52             anchors.left: parent.left
53             anchors.margins: 20
54             text: "KALİBRASYON – Sensör Seçin"
55             color: "#6b7280"
56             font.family: "Segoe UI"
57             font.pixelSize: 12
58             font.bold: true
59             font.letterSpacing: 1
60         }
61
62         Row {
63             anchors.centerIn: parent
64             spacing: 24
```



```

65
66         Rectangle {
67             id: egimKarti
68             width: 350
69             height: 450
70             radius: 12
71             color: "#0f1420"
72             border.color: egimKarti.hovered ? "#14b8a6" : "#1e2a3f"
73             border.width: 1
74             property bool hovered: false
75             scale: hovered ? 1.03 : 1.0
76             Behavior on border.color { ColorAnimation { duration: 120 } }
77             Behavior on scale { NumberAnimation { duration: 120; easing.type: Easing.OutQuad } }
78
79             Column {
80                 anchors.centerIn: parent
81                 spacing: 14
82
83                 Item {
84                     anchors.horizontalCenter: parent.horizontalCenter
85                     width: 210
86                     height: 150
87
88                     Image {
89                         anchors.centerIn: parent
90                         width: parent.width
91                         height: parent.height
92                         source: "../assets/egim_sensor.png"
93                         fillMode: Image.PreserveAspectFit
94                     }
95                 }
96
97                 Text { anchors.horizontalCenter: parent.horizontalCenter; text: "Eğim Sensörü";
color: "#dce8f5"; font.family: "Segoe UI"; font.pixelSize: 24; font.bold: true }

```

```
98         Text { anchors.horizontalCenter: parent.horizontalCenter; text: "Yatay hizalama
kalibrasyonu"; color: "#6b7280"; font.family: "Segoe UI"; font.pixelSize: 14 }

99
100        Rectangle {
101            anchors.horizontalCenter: parent.horizontalCenter
102            visible: egimKayitliVar
103            width: kaliMetni1.implicitWidth + 16
104            height: 20
105            radius: 10
106            color: "#123321"
107
108            Text {
109                id: kaliMetni1
110                anchors.centerIn: parent
111                text: "✓ Kalibre Edildi"
112                color: "#4ade80"
113                font.pixelSize: 10
114                font.bold: true
115            }
116        }
117    }
118
119    MouseArea {
120        anchors.fill: parent
121        hoverEnabled: true
122        cursorShape: Qt.PointingHandCursor
123        onEntered: egimKarti.hovered = true
124        onExited: egimKarti.hovered = false
125        onClicked: { seciliSensor = 0; egimAdimi = 0 }
126    }
127 }
128
129 Rectangle {
```

```

130         id: loadCellKarti
131
132         width: 350
133
134         height: 450
135
136         radius: 12
137
138         color: "#0f1420"
139
140         border.color: loadCellKarti.hovered ? "#3b82f6" : "#1e2a3f"
141
142         border.width: 1
143
144         property bool hovered: false
145
146         scale: hovered ? 1.03 : 1.0
147
148         Behavior on border.color { ColorAnimation { duration: 120 } }
149
150         Behavior on scale { NumberAnimation { duration: 120; easing.type: Easing.OutQuad } }
151
152
153         Column {
154
155             anchors.centerIn: parent
156
157             spacing: 14
158
159
160             Item {
161
162                 anchors.horizontalCenter: parent.horizontalCenter
163
164                 width: 210
165
166                 height: 150
167
168
169                 Image {
170
171                     anchors.centerIn: parent
172
173                     width: parent.width
174
175                     height: parent.height
176
177                     source: "../assets/loadcell.png"
178
179                     fillMode: Image.PreserveAspectFit
180
181                 }
182
183             }
184
185
186             Text { anchors.horizontalCenter: parent.horizontalCenter; text: "Load Cell";
187 color: "#dce8f5"; font.family: "Segoe UI"; font.pixelSize: 24; font.bold: true }
188
189             Text { anchors.horizontalCenter: parent.horizontalCenter; text: "Basınç sensörü
190 kalibrasyonu"; color: "#6b7280"; font.family: "Segoe UI"; font.pixelSize: 14 }

```

```
162
163         Rectangle {
164             anchors.horizontalCenter: parent.horizontalCenter
165             visible: loadCellKayitliVar
166             width: kaliMetni2.implicitWidth + 16
167             height: 20
168             radius: 10
169             color: "#123321"
170
171             Text {
172                 id: kaliMetni2
173                 anchors.centerIn: parent
174                 text: "✓ Kalibre Edildi"
175                 color: "#4ade80"
176                 font.pixelSize: 10
177                 font.bold: true
178             }
179         }
180     }
181
182     MouseArea {
183         anchors.fill: parent
184         hoverEnabled: true
185         cursorShape: Qt.PointingHandCursor
186         onEntered: loadCellKarti.hovered = true
187         onExited: loadCellKarti.hovered = false
188         onClicked: { seciliSensor = 1; loadCellAdimi = 0 }
189     }
190 }
191
192 Rectangle {
193     id: mesafeKarti
194     width: 350
```

```
195         height: 450
196         radius: 12
197         color: "#0f1420"
198         border.color: mesafeKarti.hovered ? "#9333ea" : "#1e2a3f"
199         border.width: 1
200         property bool hovered: false
201         scale: hovered ? 1.03 : 1.0
202         Behavior on border.color { ColorAnimation { duration: 120 } }
203         Behavior on scale { NumberAnimation { duration: 120; easing.type: Easing.OutQuad } }
204
205         Column {
206             anchors.centerIn: parent
207             spacing: 14
208
209             Item {
210                 anchors.horizontalCenter: parent.horizontalCenter
211                 width: 210
212                 height: 150
213
214                 Image {
215                     anchors.centerIn: parent
216                     width: parent.width
217                     height: parent.height
218                     source: "../assets/mesafe_sensor.png"
219                     fillMode: Image.PreserveAspectFit
220                 }
221             }
222
223             Text { anchors.horizontalCenter: parent.horizontalCenter; text: "Mesafe
224             Sensörü"; color: "#dce8f5"; font.family: "Segoe UI"; font.pixelSize: 24; font.bold: true }
225
226             Text { anchors.horizontalCenter: parent.horizontalCenter; text: "Konum ölçümü
227             kalibrasyonu"; color: "#6b7280"; font.family: "Segoe UI"; font.pixelSize: 14 }
228
229             Rectangle {
```

```
227         anchors.horizontalCenter: parent.horizontalCenter
228         visible: mesafeUstKayitliVar && mesafeAltKayitliVar
229         width: kaliMetni3.implicitWidth + 16
230         height: 20
231         radius: 10
232         color: "#123321"
233
234         Text {
235             id: kaliMetni3
236             anchors.centerIn: parent
237             text: "✓ Kalibre Edildi"
238             color: "#4ade80"
239             font.pixelSize: 10
240             font.bold: true
241         }
242     }
243 }
244
245 MouseArea {
246     anchors.fill: parent
247     hoverEnabled: true
248     cursorShape: Qt.PointingHandCursor
249     onEntered: mesafeKarti.hovered = true
250     onExited: mesafeKarti.hovered = false
251     onClicked: { seciliSensor = 2; mesafeAdimi = 0 }
252 }
253 }
254 }
255 }
256
257 Item {
258     Row {
259         anchors.top: parent.top
```

```
260         anchors.left: parent.left
261         anchors.margins: 20
262         spacing: 12
263
264     Button {
265         text: "← Geri"
266         font.pixelSize: 13
267
268         background: Rectangle {
269             radius: 6
270             color: parent.hovered ? "#1e2a3f" : "#0f1420"
271             border.color: "#1e2a3f"
272             border.width: 1
273             Behavior on color { ColorAnimation { duration: 120 } }
274         }
275
276         contentItem: Text {
277             text: parent.text
278             color: "#dce8f5"
279             font: parent.font
280             horizontalAlignment: Text.AlignHCenter
281             verticalAlignment: Text.AlignVCenter
282             leftPadding: 10
283             rightPadding: 10
284         }
285
286         onClicked: seciliSensor = -1
287     }
288
289     Rectangle {
290         width: 8
291         height: 8
292         radius: 4
```

```
293         anchors.verticalCenter: parent.verticalCenter
294
295         color: {
296             if (seciliSensor === 0) return "#14b8a6"
297             if (seciliSensor === 1) return "#3b82f6"
298             if (seciliSensor === 2) return "#9333ea"
299             return "#6b7280"
300         }
301
302     Text {
303         anchors.verticalCenter: parent.verticalCenter
304
305         text: {
306             if (seciliSensor === 0) return "Eğim Sensörü Kalibrasyonu"
307             if (seciliSensor === 1) return "Load Cell Kalibrasyonu"
308             if (seciliSensor === 2) return "Mesafe Sensörü Kalibrasyonu"
309             return ""
310         }
311
312         color: "#dce8f5"
313
314         font.family: "Segoe UI"
315
316         font.pixelSize: 16
317
318         font.bold: true
319     }
320 }
321
322 Item {
323     anchors.top: parent.top
324
325     anchors.left: parent.left
326
327     anchors.right: parent.right
328
329     anchors.bottom: parent.bottom
330
331     anchors.topMargin: 70
332
333     visible: seciliSensor === 0
334
335     Column {
```



```
326         anchors.centerIn: parent
327         spacing: 20
328         width: 360
329
330         Rectangle {
331             anchors.horizontalCenter: parent.horizontalCenter
332             visible: egimKayitliVar
333             width: sonKaliMetni1.implicitWidth + 20
334             height: 26
335             radius: 13
336             color: "#0f1420"
337             border.color: "#1e2a3f"
338             border.width: 1
339
340             Text {
341                 id: sonKaliMetni1
342                 anchors.centerIn: parent
343                 text: "Son kalibrasyon: " + egimSonTarih
344                 color: "#6b7280"
345                 font.pixelSize: 11
346             }
347         }
348
349         Text {
350             width: parent.width
351             horizontalAlignment: Text.AlignHCenter
352             wrapMode: Text.WordWrap
353             text: "Cihazı düz ve sabit bir yüzeye yerleştirin"
354             color: "#dce8f5"
355             font.family: "Segoe UI"
356             font.pixelSize: 14
357         }
358
```

```
359         SuTerazisi {
360             anchors.horizontalCenter: parent.horizontalCenter
361             egimX: sensorManager.egimX
362             egimY: sensorManager.egimY
363         }
364
365         Rectangle {
366             anchors.horizontalCenter: parent.horizontalCenter
367             width: 220
368             height: 72
369             radius: 12
370             color: "#0f2e2a"
371             border.color: "#14b8a6"
372             border.width: 1
373
374             Text {
375                 anchors.centerIn: parent
376                 text: "X: " + sensorManager.egimX.toFixed(1) + "°   Y: " +
sensorManager.egimY.toFixed(1) + "°"
377                 color: "#2dd4bf"
378                 font.family: "Segoe UI"
379                 font.pixelSize: 22
380                 font.bold: true
381             }
382         }
383
384         Button {
385             width: parent.width
386             height: 44
387             text: "Sıfır1a (Zero)"
388             font.pixelSize: 14
389             font.bold: true
390
391             onClicked: {
```

```
392         var eskiKal = database.kalibrasyonGetir("egim")
393         var eskiOffsetX = eskiKal.mevcut ? eskiKal.deger1 : 0.0
394         var eskiOffsetY = eskiKal.mevcut ? eskiKal.deger2 : 0.0
395         var yeniOffsetX = eskiOffsetX + sensorManager.egimX
396         var yeniOffsetY = eskiOffsetY + sensorManager.egimY
397         database.kalibrasyonKaydet("egim", yeniOffsetX, yeniOffsetY)
398         console.log("Egim sensoru kalibrasyonu kaydedildi.")
399         seciliSensor = -1
400     }
401
402     background: Rectangle {
403         radius: 8
404         color: parent.hovered ? "#4f8cf7" : "#3b82f6"
405         Behavior on color { ColorAnimation { duration: 120 } }
406     }
407
408     contentItem: Text {
409         text: parent.text
410         color: "#ffffff"
411         font: parent.font
412         horizontalAlignment: Text.AlignHCenter
413         verticalAlignment: Text.AlignVCenter
414     }
415 }
416 }
417 }
418
419 Item {
420     anchors.top: parent.top
421     anchors.left: parent.left
422     anchors.right: parent.right
423     anchors.bottom: parent.bottom
424     anchors.topMargin: 70
```

```

425         visible: seciliSensor === 1
426
427         Column {
428             anchors.centerIn: parent
429             spacing: 20
430             width: 360
431
432             Rectangle {
433                 anchors.horizontalCenter: parent.horizontalCenter
434                 visible: loadCellKayitliVar
435                 width: sonKaliMetni2.implicitWidth + 20
436                 height: 26
437                 radius: 13
438                 color: "#0f1420"
439                 border.color: "#1e2a3f"
440                 border.width: 1
441
442                 Text {
443                     id: sonKaliMetni2
444                     anchors.centerIn: parent
445                     text: "Son referans: " + loadCellReferans.toFixed(1) + " kg - " +
loadCellSonTarih
446                     color: "#6b7280"
447                     font.pixelSize: 11
448                 }
449             }
450
451             Text {
452                 anchors.horizontalCenter: parent.horizontalCenter
453                 text: "Adım " + (loadCellAdimi + 1) + " / 2"
454                 color: "#6b7280"
455                 font.family: "Segoe UI"
456                 font.pixelSize: 11

```

```
457         font.letterSpacing: 1
458     }
459
460     Row {
461         anchors.horizontalCenter: parent.horizontalCenter
462         spacing: 6
463
464         Repeater {
465             model: 2
466
467             Rectangle {
468                 width: index === loadCellAdimi ? 22 : 8
469                 height: 8
470                 radius: 4
471                 color: index <= loadCellAdimi ? "#3b82f6" : "#1e2a3f"
472                 Behavior on width { NumberAnimation { duration: 150 } }
473             }
474         }
475
476         Column {
477             width: parent.width
478             spacing: 20
479             visible: loadCellAdimi === 0
480
481             Text {
482                 width: parent.width
483                 horizontalAlignment: Text.AlignHCenter
484                 wrapMode: Text.WordWrap
485                 text: "Load cell üzerinde hiçbir ağırlık olmadığından emin olun"
486                 color: "#dce8f5"
487                 font.family: "Segoe UI"
488                 font.pixelSize: 14
489             }
```

```
490
491         Rectangle {
492             anchors.horizontalCenter: parent.horizontalCenter
493             width: 220
494             height: 72
495             radius: 12
496             color: "#132335"
497             border.color: "#3b82f6"
498             border.width: 1
499
500         Text {
501             anchors.centerIn: parent
502             text: "Ham deger: " + sensorManager.hamAgirlik.toFixed(0)
503             color: "#3b82f6"
504             font.family: "Segoe UI"
505             font.pixelSize: 22
506             font.bold: true
507         }
508     }
509
510     Button {
511         width: parent.width
512         height: 44
513         text: "Sıfırla ve Devam Et"
514         font.pixelSize: 14
515         font.bold: true
516
517         background: Rectangle {
518             radius: 8
519             color: parent.hovered ? "#4f8cf7" : "#3b82f6"
520             Behavior on color { ColorAnimation { duration: 120 } }
521         }
522
```

```
523         contentItem: Text {
524             text: parent.text
525             color: "#ffffff"
526             font: parent.font
527             horizontalAlignment: Text.AlignHCenter
528             verticalAlignment: Text.AlignVCenter
529         }
530
531         onClicked: {
532             loadCellSifirHam = sensorManager.hamAgirlik
533             loadCellAdimi = 1
534         }
535     }
536 }
537
538 Column {
539     width: parent.width
540     spacing: 20
541     visible: loadCellAdimi === 1
542
543     Text {
544         width: parent.width
545         horizontalAlignment: Text.AlignHCenter
546         wrapMode: Text.WordWrap
547         text: "Bilinen bir referans ağırlığı yerleştirin"
548         color: "#dce8f5"
549         font.family: "Segoe UI"
550         font.pixelSize: 14
551     }
552
553     Row {
554         anchors.horizontalCenter: parent.horizontalCenter
555         spacing: 12
```

```

556
557         Button {
558             width: 100
559             height: 40
560             text: "1.6 kg"
561             font.pixelSize: 13
562
563             onClicked: {
564                 var fark = sensorManager.hamAgirlik - loadCellSifirHam
565                 var katsayi = fark !== 0 ? fark / (1.6 * 1000) : 2280.0
566                 database.kalibrasyonKaydet("loadcell", katsayi, loadCellSifirHam)
567                 console.log("Load cell kalibrasyonu kaydedildi (1.6kg referans).")
568                 seciliSensor = -1
569             }
570
571             background: Rectangle {
572                 radius: 8
573                 color: parent.hovered ? "#1e2a3f" : "#0a0e17"
574                 border.color: "#1e2a3f"
575                 border.width: 1
576                 Behavior on color { ColorAnimation { duration: 120 } }
577             }
578
579             contentItem: Text {
580                 text: parent.text
581                 color: "#dce8f5"
582                 font: parent.font
583                 horizontalAlignment: Text.AlignHCenter
584                 verticalAlignment: Text.AlignVCenter
585             }
586         }
587
588         Button {

```



```

589         width: 100
590         height: 40
591         text: "4.8 kg"
592         font.pixelSize: 13
593
594         onClicked: {
595             var fark = sensorManager.hamAgirlik - loadCellSifirHam
596             var katsayi = fark != 0 ? fark / (4.8 * 1000) : 2280.0
597             database.kalibrasyonKaydet("loadcell", katsayi, loadCellSifirHam)
598             console.log("Load cell kalibrasyonu kaydedildi (4.8kg referans).")
599             seciliSensor = -1
600         }
601
602         background: Rectangle {
603             radius: 8
604             color: parent.hovered ? "#1e2a3f" : "#0a0e17"
605             border.color: "#1e2a3f"
606             border.width: 1
607             Behavior on color { ColorAnimation { duration: 120 } }
608         }
609
610         contentItem: Text {
611             text: parent.text
612             color: "#dce8f5"
613             font: parent.font
614             horizontalAlignment: Text.AlignHCenter
615             verticalAlignment: Text.AlignVCenter
616         }
617     }
618 }
619 }
620 }
621 }

```

```
622
623     Item {
624         anchors.top: parent.top
625         anchors.left: parent.left
626         anchors.right: parent.right
627         anchors.bottom: parent.bottom
628         anchors.topMargin: 70
629         visible: seciliSensor === 2
630
631         Column {
632             anchors.centerIn: parent
633             spacing: 20
634             width: 360
635
636             Rectangle {
637                 anchors.horizontalCenter: parent.horizontalCenter
638                 visible: mesafeUstKayitliVar || mesafeAltKayitliVar
639                 width: sonKaliMetni3.implicitWidth + 20
640                 height: 26
641                 radius: 13
642                 color: "#0f1420"
643                 border.color: "#1e2a3f"
644                 border.width: 1
645
646                 Text {
647                     id: sonKaliMetni3
648                     anchors.centerIn: parent
649                     text: "Son kalibrasyon: " + mesafeSonTarih
650                     color: "#6b7280"
651                     font.pixelSize: 11
652                 }
653             }
654
```

```

655         Text {
656             anchors.horizontalCenter: parent.horizontalCenter
657             text: "Adım " + (mesafeAdimi + 1) + " / 2"
658             color: "#6b7280"
659             font.family: "Segoe UI"
660             font.pixelSize: 11
661             font.letterSpacing: 1
662         }
663
664     Row {
665         anchors.horizontalCenter: parent.horizontalCenter
666         spacing: 6
667
668         Repeater {
669             model: 2
670
671             Rectangle {
672                 width: index === mesafeAdimi ? 22 : 8
673                 height: 8
674                 radius: 4
675                 color: index <= mesafeAdimi ? "#9333ea" : "#1e2a3f"
676                 Behavior on width { NumberAnimation { duration: 150 } }
677             }
678         }
679
680     Column {
681         width: parent.width
682         spacing: 20
683         visible: mesafeAdimi === 0
684
685         Text {
686             width: parent.width
687             horizontalAlignment: Text.AlignHCenter

```

```
688         wrapMode: Text.WordWrap
689         text: "Boruyu üst pozisyona getirin ve sabitleyin"
690         color: "#dce8f5"
691         font.family: "Segoe UI"
692         font.pixelSize: 14
693     }
694
695     Rectangle {
696         anchors.horizontalCenter: parent.horizontalCenter
697         width: 220
698         height: 72
699         radius: 12
700         color: "#241a38"
701         border.color: "#9333ea"
702         border.width: 1
703
704         Text {
705             anchors.centerIn: parent
706             text: sensorManager.konum.toFixed(1) + " mm"
707             color: "#c084fc"
708             font.family: "Segoe UI"
709             font.pixelSize: 22
710             font.bold: true
711         }
712     }
713
714     Button {
715         width: parent.width
716         height: 44
717         text: "Üst Pozisyonu Kaydet"
718         font.pixelSize: 14
719         font.bold: true
720     }
```

```

721         onClicked: {
722             database.kalibrasyonKaydet("mesafe_ust", sensorManager.konum, 0.0)
723             mesafeAdimi = 1
724         }
725
726         background: Rectangle {
727             radius: 8
728             color: parent.hovered ? "#4f8cf7" : "#3b82f6"
729             Behavior on color { ColorAnimation { duration: 120 } }
730         }
731
732         contentItem: Text {
733             text: parent.text
734             color: "#ffffff"
735             font: parent.font
736             horizontalAlignment: Text.AlignHCenter
737             verticalAlignment: Text.AlignVCenter
738         }
739     }
740 }
741
742 Column {
743     width: parent.width
744     spacing: 20
745     visible: mesafeAdimi === 1
746
747     Text {
748         width: parent.width
749         horizontalAlignment: Text.AlignHCenter
750         wrapMode: Text.WordWrap
751         text: "Boruyu alt pozisyona getirin ve sabitleyin"
752         color: "#dce8f5"
753         font.family: "Segoe UI"

```

```
754         font.pixelSize: 14
755     }
756
757     Rectangle {
758         anchors.horizontalCenter: parent.horizontalCenter
759         width: 220
760         height: 72
761         radius: 12
762         color: "#241a38"
763         border.color: "#9333ea"
764         border.width: 1
765
766         Text {
767             anchors.centerIn: parent
768             text: sensorManager.konum.toFixed(1) + " mm"
769             color: "#c084fc"
770             font.family: "Segoe UI"
771             font.pixelSize: 22
772             font.bold: true
773         }
774     }
775
776     Button {
777         width: parent.width
778         height: 44
779         text: "Alt Pozisyonu Kaydet ve Tamamla"
780         font.pixelSize: 14
781         font.bold: true
782
783         onClicked: {
784             database.kalibrasyonKaydet("mesafe_alt", sensorManager.konum, 0.0)
785             seciliSensor = -1
786         }
787     }
788 }
```

```
787
788         background: Rectangle {
789             radius: 8
790             color: parent.hovered ? "#22c55e" : "#16a34a"
791             Behavior on color { ColorAnimation { duration: 120 } }
792         }
793
794         contentItem: Text {
795             text: parent.text
796             color: "#ffffff"
797             font: parent.font
798             horizontalAlignment: Text.AlignHCenter
799             verticalAlignment: Text.AlignVCenter
800         }
801     }
802 }
803 }
804 }
805 }
806 }
807 }
```

```
1 import QtQuick 6.7
2 import QtQuick.Controls 6.7
3
4 Rectangle {
5     id: sayfa
6
7     property bool sifreGorunur: false
8     property bool beniHatirla: false
9
10    signal girisBasarili()
11
12    gradient: Gradient {
13        GradientStop { position: 0.0; color: "#101c33" }
14        GradientStop { position: 1.0; color: "#0a1220" }
15    }
16
17    Row {
18        anchors.top: parent.top
19        anchors.right: parent.right
20        anchors.margins: 24
21        spacing: 8
22
23        Rectangle {
24            width: 44
25            height: 30
26            radius: 6
27            color: "#0f1420"
28            border.color: "#3b82f6"
29            border.width: 1
30
31            Text {
```



```
32         anchors.centerIn: parent
33         text: "TR"
34         color: "#ffffff"
35         font.pixelSize: 12
36         font.bold: true
37     }
38 }
39
40 Rectangle {
41     width: 44
42     height: 30
43     radius: 6
44     color: "#0f1420"
45     border.color: "#1e2a3f"
46     border.width: 1
47
48     Text {
49         anchors.centerIn: parent
50         text: "EN"
51         color: "#6b7280"
52         font.pixelSize: 12
53     }
54 }
55 }
56
57 Rectangle {
58     anchors.centerIn: kart
59     width: kart.width + 48
60     height: kart.height + 48
61     radius: 28
62     color: "#18000000"
63 }
64
```

```
65     Rectangle {
66         anchors.centerIn: kart
67         width: kart.width + 32
68         height: kart.height + 32
69         radius: 24
70         color: "#25000000"
71     }
72
73     Rectangle {
74         anchors.centerIn: kart
75         width: kart.width + 18
76         height: kart.height + 18
77         radius: 20
78         color: "#35000000"
79     }
80
81     Rectangle {
82         anchors.centerIn: kart
83         width: kart.width + 8
84         height: kart.height + 8
85         radius: 18
86         color: "#45000000"
87     }
88
89     Rectangle {
90         id: kart
91         anchors.centerIn: parent
92         width: 400
93         color: "#0f1420"
94         radius: 16
95         border.color: "#4b5563"
96         border.width: 1
97         height: govde.implicitHeight + 48
```

```
98
99     Column {
100         id: govde
101         anchors.top: parent.top
102         anchors.horizontalCenter: parent.horizontalCenter
103         anchors.topMargin: 32
104         width: parent.width - 48
105         spacing: 20
106
107         Image {
108             anchors.horizontalCenter: parent.horizontalCenter
109             source: "../assets/logo.png"
110             width: 144
111             height: 144
112             fillMode: Image.PreserveAspectFit
113         }
114
115         Rectangle {
116             width: parent.width
117             height: 1
118             color: "#1e2a3f"
119         }
120
121         Column {
122             width: parent.width
123             spacing: 4
124
125             Text {
126                 anchors.horizontalCenter: parent.horizontalCenter
127                 text: "Sisteme Giriş"
128                 color: "ffffff"
129                 font.family: "Segoe UI"
130                 font.pixelSize: 22
```

```
131         font.bold: true
132     }
133
134     Text {
135         anchors.horizontalCenter: parent.horizontalCenter
136         text: "Yetkili hesabınızla devam edin"
137         color: "#6b7280"
138         font.family: "Segoe UI"
139         font.pixelSize: 12
140     }
141 }
142
143 Column {
144     width: parent.width
145     spacing: 6
146
147     Text {
148         text: "KULLANICI ADI"
149         color: "#6b7280"
150         font.family: "Segoe UI"
151         font.pixelSize: 10
152         font.letterSpacing: 1
153     }
154
155     TextField {
156         id: kullaniciAdiKutusu
157         width: parent.width
158         height: 42
159         placeholderText: "kullanici_adi"
160         placeholderTextColor: "#4b5563"
161         color: "ffffff"
162         font.pixelSize: 13
163         leftPadding: 12
```

```
164         verticalAlignment: TextInput.AlignVCenter
165         background: Rectangle {
166             color: "#0a0e17"
167             radius: 6
168             border.color: kullaniciAdiKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
169             border.width: 1
170         }
171     }
172 }
173
174 Column {
175     width: parent.width
176     spacing: 6
177
178     Text {
179         text: "ŞİFRE"
180         color: "#6b7280"
181         font.family: "Segoe UI"
182         font.pixelSize: 10
183         font.letterSpacing: 1
184     }
185
186     TextField {
187         id: sifreKutusu
188         width: parent.width
189         height: 42
190         placeholderText: "*****"
191         placeholderTextColor: "#4b5563"
192         echoMode: sayfa.sifreGorunur ? TextInput.Normal : TextInput.Password
193         color: "ffffff"
194         font.pixelSize: 13
195         leftPadding: 12
196         rightPadding: 48
```

```
197         verticalAlignment: TextInput.AlignVCenter
198         background: Rectangle {
199             color: "#0a0e17"
200             radius: 6
201             border.color: sifreKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
202             border.width: 1
203         }
204
205         Rectangle {
206             id: sifreGosterButonu
207             anchors.right: parent.right
208             anchors.verticalCenter: parent.verticalCenter
209             anchors.rightMargin: 8
210             width: 28
211             height: 28
212             radius: 6
213             color: sifreGosterMouse.pressed ? "#162033" : (sifreGosterMouse.containsMouse ?
"#111827" : "transparent")
214             border.color: sayfa.sifreGorunur ? "#334155" : "transparent"
215             border.width: 1
216
217             property color ikonRengi: sifreGosterMouse.containsMouse || sayfa.sifreGorunur ?
"#cbd5e1" : "#6b7280"
218
219             Canvas {
220                 id: sifreIkonu
221                 anchors.centerIn: parent
222                 width: 18
223                 height: 18
224
225                 onPaint: {
226                     var ctx = getContext("2d")
227                     ctx.clearRect(0, 0, width, height)
228                     ctx.lineWidth = 1.7
```

```
229         ctx.lineCap = "round"
230         ctx.lineJoin = "round"
231         ctx.strokeStyle = sifreGosterButonu.ikonRengi
232
233         ctx.beginPath()
234         ctx.moveTo(1.5, 9)
235         ctx.bezierCurveTo(4.2, 4.8, 7.0, 3.8, 9, 3.8)
236         ctx.bezierCurveTo(11.0, 3.8, 13.8, 4.8, 16.5, 9)
237         ctx.bezierCurveTo(13.8, 13.2, 11.0, 14.2, 9, 14.2)
238         ctx.bezierCurveTo(7.0, 14.2, 4.2, 13.2, 1.5, 9)
239         ctx.stroke()
240
241         ctx.beginPath()
242         ctx.arc(9, 9, 2.5, 0, Math.PI * 2, false)
243         ctx.stroke()
244
245         if (sayfa.sifreGorunur) {
246             ctx.beginPath()
247             ctx.moveTo(3, 15)
248             ctx.lineTo(15, 3)
249             ctx.stroke()
250         }
251     }
252
253     Connections {
254         target: sayfa
255         function onSifreGorunurChanged() {
256             sifreIkonu.requestPaint()
257         }
258     }
259
260     Connections {
261         target: sifreGosterButonu
```

```

262         function onIkonRengiChanged() {
263             sifreIkonu.requestPaint()
264         }
265     }
266 }
267
268 MouseArea {
269     id: sifreGosterMouse
270     anchors.fill: parent
271     hoverEnabled: true
272     cursorShape: Qt.PointingHandCursor
273     onClicked: sayfa.sifreGorunur = !sayfa.sifreGorunur
274 }
275 }
276
277 Text {
278     anchors.right: parent.right
279     anchors.verticalCenter: parent.verticalCenter
280     anchors.rightMargin: 12
281     text: sayfa.sifreGorunur ? "👁" : "👁"
282     color: "#6b7280"
283     font.pixelSize: 14
284     visible: false
285
286     MouseArea {
287         anchors.fill: parent
288         anchors.margins: -8
289         cursorShape: Qt.PointingHandCursor
290         onClicked: sayfa.sifreGorunur = !sayfa.sifreGorunur
291     }
292 }
293 }
294 }

```



```
295
296     Button {
297         width: parent.width
298         height: 44
299         text: "Giriş Yap"
300         font.family: "Segoe UI"
301         font.pixelSize: 14
302         font.bold: true
303
304         onClicked: {
305             var sonuc = backend.girisYap(kullaniciAdiKutusu.text, sifreKutusu.text)
306
307             if (sonuc) {
308                 sayfa.girisBasarili()
309             } else {
310                 console.log("Kullanici adi veya sifre hatali")
311             }
312         }
313
314         background: Rectangle {
315             radius: 8
316             color: parent.pressed ? "#2563eb" : (parent.hovered ? "#4f8cf7" : "#3b82f6")
317             Behavior on color { ColorAnimation { duration: 120 } }
318         }
319
320         contentItem: Text {
321             text: parent.text
322             color: "ffffff"
323             font: parent.font
324             horizontalAlignment: Text.AlignHCenter
325             verticalAlignment: Text.AlignVCenter
326         }
327     }
```

```
328
329     Rectangle {
330         width: parent.width
331         height: 1
332         color: "#1e2a3f"
333     }
334
335     Text {
336         anchors.horizontalCenter: parent.horizontalCenter
337         text: "\u00A9 2026 Liya Test A.Ş. SLIPER Analiz Yazılımı"
338         color: "#4b5563"
339         font.family: "Segoe UI"
340         font.pixelSize: 10
341     }
342 }
343 }
344 }
```

```
1 import QtQuick 6.7
2 import QtQuick.Controls 6.7
3 import QtCharts 6.7
4
5 Rectangle {
6     color: "#0a0e17"
7     property real zamanSayaci: 0
8     property int aktifOlcumId: -1
9     property string uyariMesaji: ""
10
11     property var basincGecmis: []
12     property var konumGecmis: []
13     property var hizGecmis: []
14     property var debiGecmis: []
15
16     signal olcumTamamlandi(int id)
17
18     function eksenTavaniHesapla(dizi, tavan, taban) {
19         if (dizi.length === 0) return taban
20         var maxDeger = Math.max.apply(null, dizi)
21         var hedef = maxDeger * 1.25
22         if (hedef < taban) hedef = taban
23         if (hedef > tavan) hedef = tavan
24         return hedef
25     }
26
27     Connections {
28         target: sensorManager
29         function onVeriGuncellendi() {
30             if (!sensorManager.veriGecerli || aktifOlcumId <= 0 || calculator.duraklatildi) return
31
```

```
32         zamanSayaci += 0.2
33
34         basincSerisi.append(zamanSayaci, sensorManager.basinc)
35         konumSerisi.append(zamanSayaci, sensorManager.konum)
36         hizSerisi.append(zamanSayaci, sensorManager.hiz)
37         debiSerisi.append(zamanSayaci, sensorManager.debi)
38
39         basincGecmis.push(sensorManager.basinc)
40         konumGecmis.push(sensorManager.konum)
41         hizGecmis.push(sensorManager.hiz)
42         debiGecmis.push(sensorManager.debi)
43
44         if (zamanSayaci > 60) {
45             basincSerisi.remove(0)
46             konumSerisi.remove(0)
47             hizSerisi.remove(0)
48             debiSerisi.remove(0)
49
50             basincGecmis.shift()
51             konumGecmis.shift()
52             hizGecmis.shift()
53             debiGecmis.shift()
54
55             xEkseni.min = zamanSayaci - 60
56             xEkseni.max = zamanSayaci
57             konumXEkseni.min = zamanSayaci - 60
58             konumXEkseni.max = zamanSayaci
59             hizXEkseni.min = zamanSayaci - 60
60             hizXEkseni.max = zamanSayaci
61             debiXEkseni.min = zamanSayaci - 60
62             debiXEkseni.max = zamanSayaci
63         }
64
```

```
65         yEkseni.max = eksenTavaniHesapla(basincGecmis, 300, 20)
66         konumYEkseni.max = eksenTavaniHesapla(konumGecmis, 500, 50)
67         hizYEkseni.max = eksenTavaniHesapla(hizGecmis, 4, 0.5)
68         debiYEkseni.max = eksenTavaniHesapla(debiGecmis, 180, 10)
69
70         calculator.konumGuncelle(sensorManager.konum)
71     }
72 }
73
74 Row {
75     anchors.fill: parent
76     spacing: 0
77
78     Rectangle {
79         id: testBilgileriPaneli
80         width: 280
81         height: parent.height
82         color: "#0f1420"
83
84         Text {
85             anchors.top: parent.top
86             anchors.left: parent.left
87             anchors.margins: 20
88             text: "TEST BİLGİLERİ"
89             color: "#6b7280"
90             font.family: "Segoe UI"
91             font.pixelSize: 12
92             font.bold: true
93             font.letterSpacing: 1
94         }
95
96         Column {
97             id: formAlanlari
```

```
98         anchors.top: parent.top
99         anchors.left: parent.left
100        anchors.right: parent.right
101        anchors.margins: 20
102        anchors.topMargin: 60
103        spacing: 6
104
105        Text {
106            text: "Müşteri"
107            color: "#9ca3af"
108            font.family: "Segoe UI"
109            font.pixelSize: 12
110        }
111
112        TextField {
113            id: musterikutusu
114            width: parent.width
115            height: 38
116            placeholderText: "Musteri adi girin..."
117            placeholderTextColor: "#4b5563"
118            color: "ffffff"
119            font.pixelSize: 13
120            leftPadding: 10
121            verticalAlignment: TextInput.AlignVCenter
122            background: Rectangle {
123                color: "#0a0e17"
124                radius: 6
125                border.color: musterikutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
126                border.width: 1
127            }
128        }
129
130        Text {
```

```

131         text: "Beton Reçetesi"
132         color: "#9ca3af"
133         font.family: "Segoe UI"
134         font.pixelSize: 12
135     }
136
137     ComboBox {
138         id: receteKutusu
139         width: parent.width
140         height: 38
141         model: ["Reçete Seçin...", "C25/30 Standart", "C30/37 SCC", "C35/45 Yüksek
Mukavemet"]
142
143         background: Rectangle {
144             color: "#0a0e17"
145             radius: 6
146             border.color: receteKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
147             border.width: 1
148         }
149
150         contentItem: Text {
151             text: receteKutusu.displayText
152             color: "#9ca3af"
153             font.pixelSize: 13
154             leftPadding: 10
155             verticalAlignment: Text.AlignVCenter
156         }
157     }
158
159     Text {
160         text: "Eklenen Ağırlık (kg)"
161         color: "#9ca3af"
162         font.family: "Segoe UI"
163         font.pixelSize: 12

```

```
164         }
165
166         TextField {
167             id: agirlikKutusu
168             width: parent.width
169             height: 38
170             placeholderText: "örn: 4.8"
171             placeholderTextColor: "#4b5563"
172             color: "#ffffff"
173             font.pixelSize: 13
174             leftPadding: 10
175             verticalAlignment: TextInput.AlignVCenter
176             validator: DoubleValidator { bottom: 0; top: 20; decimals: 1 }
177             background: Rectangle {
178                 color: "#0a0e17"
179                 radius: 6
180                 border.color: agirlikKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
181                 border.width: 1
182             }
183         }
184
185         Item { width: parent.width; height: 10 }
186
187         Button {
188             id: baslatButonu
189             width: parent.width
190             height: 44
191             text: "▶ Ölçümü Başlat"
192             font.family: "Segoe UI"
193             font.pixelSize: 14
194             font.bold: true
195             enabled: aktifOlcumId <= 0
196
```



```
197         onClicked: {
198             if (!sensorManager.veriGecerli) {
199                 console.warn("Gerçek sensor verisi yok, ölçüm baslatılmadı.")
200                 uyarıMesajı = "Cihaz bağlı değil. Lütfen önce sol alttan SLIPER-ESP32'ye
bağlanın."
201                 return
202             }
203
204             uyarıMesajı = ""
205             zamanSayacı = 0
206             basınçSerisi.clear()
207             konumSerisi.clear()
208             hızSerisi.clear()
209             debiSerisi.clear()
210             basınçGecmis = []
211             konumGecmis = []
212             hızGecmis = []
213             debiGecmis = []
214             xEksenı.min = 0
215             xEksenı.max = 60
216             yEksenı.max = 300
217             konumXEksenı.min = 0
218             konumXEksenı.max = 60
219             konumYEksenı.max = 500
220             hızXEksenı.min = 0
221             hızXEksenı.max = 60
222             hızYEksenı.max = 4
223             debiXEksenı.min = 0
224             debiXEksenı.max = 60
225             debiYEksenı.max = 180
226
227             calculator.sıfırla()
228             aktifÖlçümId = database.ölçümBaslat(
```

```

229         musterikutusu.text,
230         receteKutusu.currentText,
231         parseFloat(agirlikKutusu.text)
232     )
233     console.log("Aktif olcum id:", aktifOlcumId)
234 }
235
236     background: Rectangle {
237         radius: 8
238         color: !baslatButonu.enabled ? "#14532d" : (baslatButonu.pressed ? "#15803d" :
(baslatButonu.hovered ? "#22c55e" : "#16a34a"))
239         opacity: baslatButonu.enabled ? 1.0 : 0.5
240         Behavior on color { ColorAnimation { duration: 120 } }
241     }
242
243     contentItem: Text {
244         text: baslatButonu.text
245         color: "#ffffff"
246         opacity: baslatButonu.enabled ? 1.0 : 0.6
247         font: baslatButonu.font
248         horizontalAlignment: Text.AlignHCenter
249         verticalAlignment: Text.AlignVCenter
250     }
251 }
252
253     Text {
254         width: parent.width
255         visible: uyariMesaji.length > 0
256         text: uyariMesaji
257         color: "#f87171"
258         font.family: "Segoe UI"
259         font.pixelSize: 11
260         wrapMode: Text.WordWrap
261     }

```

```

262
263         Row {
264             width: parent.width
265             spacing: 10
266
267             Button {
268                 id: duraklatButonu
269                 width: (parent.width - parent.spacing) / 2
270                 height: 40
271                 text: calculator.duraklatildi ? "► Devam Et" : "■ Duraklat"
272                 font.family: "Segoe UI"
273                 font.pixelSize: 13
274
275                 onClicked: {
276                     if (calculator.duraklatildi) {
277                         calculator.devamEt()
278                     } else {
279                         calculator.duraklat()
280                     }
281                 }
282
283                 background: Rectangle {
284                     radius: 8
285                     color: duraklatButonu.pressed ? "#78350f" : (duraklatButonu.hovered ?
"#a16207" : "#92400e")
286                     border.color: "#d97706"
287                     border.width: 1
288                 }
289
290                 contentItem: Text {
291                     text: duraklatButonu.text
292                     color: "#ffffff"
293                     font: duraklatButonu.font

```

```
294         horizontalAlignment: Text.AlignHCenter
295         verticalAlignment: Text.AlignVCenter
296     }
297 }
298
299 Button {
300     id: bitirButonu
301     width: (parent.width - parent.spacing) / 2
302     height: 40
303     text: "■ Bitir"
304     font.family: "Segoe UI"
305     font.pixelSize: 13
306
307     onClicked: {
308         if (aktifOlcumId > 0) {
309             bitirOnayPopup.open()
310         } else {
311             uyarıMesajı = "Aktif bir ölçüm yok. Önce ölçümü başlatın."
312         }
313     }
314
315     background: Rectangle {
316         radius: 8
317         color: bitirButonu.pressed ? "#7f1d1d" : (bitirButonu.hovered ? "#b91c1c" :
"#991b1b")
318         border.color: "#dc2626"
319         border.width: 1
320     }
321
322     contentItem: Text {
323         text: bitirButonu.text
324         color: "ffffff"
325         font: bitirButonu.font
326         horizontalAlignment: Text.AlignHCenter
```

```
327         verticalAlignment: Text.AlignVCenter
328     }
329 }
330 }
331 }
332
333 Rectangle {
334     id: durumKarti
335     anchors.top: formAlanlari.bottom
336     anchors.left: parent.left
337     anchors.right: parent.right
338     anchors.topMargin: 20
339     anchors.leftMargin: 20
340     anchors.rightMargin: 20
341     height: 52
342     radius: 10
343     color: "#0a0e17"
344     border.color: "#1e2a3f"
345     border.width: 1
346
347     Row {
348         anchors.left: parent.left
349         anchors.verticalCenter: parent.verticalCenter
350         anchors.leftMargin: 14
351         spacing: 10
352
353         Rectangle {
354             id: trafikIsigi
355             width: 12
356             height: 12
357             radius: 6
358             anchors.verticalCenter: parent.verticalCenter
359             color: {
```

```
360         if (!sensorManager.veriGecerli) return "#dc2626"
361         if (calculator.duraklatildi) return "#6b7280"
362         if (calculator.durum === "YUKARIDA") return "#16a34a"
363         if (calculator.durum === "INIYOR") return "#f59e0b"
364         return "#dc2626"
365     }
366
367     Rectangle {
368         anchors.centerIn: parent
369         width: parent.width + 8
370         height: parent.height + 8
371         radius: width / 2
372         color: "transparent"
373         border.color: parent.color
374         border.width: 1
375         opacity: 0.35
376     }
377 }
378
379 Text {
380     anchors.verticalCenter: parent.verticalCenter
381     text: {
382         if (!sensorManager.veriGecerli) return "VERI YOK"
383         if (calculator.duraklatildi) return "DURAKLATILDI"
384         return calculator.durum
385     }
386     color: "#dce8f5"
387     font.family: "Segoe UI"
388     font.pixelSize: 13
389     font.bold: true
390 }
391 }
392
```

```
393         Row {
394             anchors.right: parent.right
395             anchors.verticalCenter: parent.verticalCenter
396             anchors.rightMargin: 14
397             spacing: 6
398
399             Text {
400                 anchors.verticalCenter: parent.verticalCenter
401                 text: "STROKE"
402                 color: "#6b7280"
403                 font.family: "Segoe UI"
404                 font.pixelSize: 10
405                 font.letterSpacing: 1
406             }
407
408             Text {
409                 anchors.verticalCenter: parent.verticalCenter
410                 text: calculator.strokeSayisi
411                 color: "#e8a020"
412                 font.family: "Segoe UI"
413                 font.pixelSize: 16
414                 font.bold: true
415             }
416         }
417     }
418
419     Text {
420         id: anlikDegerlerBaslik
421         anchors.top: durumKarti.bottom
422         anchors.left: parent.left
423         anchors.topMargin: 24
424         anchors.leftMargin: 20
425         text: "ANLIK DEĞERLER"
```

```
426         color: "#6b7280"
427         font.family: "Segoe UI"
428         font.pixelSize: 12
429         font.bold: true
430         font.letterSpacing: 1
431     }
432
433     Grid {
434         anchors.top: anlikDegerlerBaslik.bottom
435         anchors.left: parent.left
436         anchors.topMargin: 12
437         anchors.leftMargin: 20
438         columns: 2
439         spacing: 10
440
441         Rectangle {
442             id: basincKarti
443             width: 115
444             height: 90
445             radius: 10
446             color: "#0a0e17"
447             border.color: "#1e2a3f"
448             border.width: 1
449             clip: true
450
451             Rectangle { width: 4; height: parent.height; color: "#3b82f6" }
452
453             Column {
454                 anchors.top: parent.top
455                 anchors.right: parent.right
456                 anchors.bottom: parent.bottom
457                 anchors.left: parent.left
458                 anchors.topMargin: 10
```



```
459         anchors.rightMargin: 10
460         anchors.bottomMargin: 10
461         anchors.leftMargin: 16
462         spacing: 6
463
464         Text {
465             text: "BASINÇ"
466             color: "#6b7280"
467             font.family: "Segoe UI"
468             font.pixelSize: 10
469             font.letterSpacing: 1
470         }
471
472         Text {
473             text: sensorManager.veriGecerli ? sensorManager.basinc.toFixed(1) : "--"
474             color: "#3b82f6"
475             font.family: "Segoe UI"
476             font.pixelSize: 22
477             font.bold: true
478         }
479
480         Text {
481             text: "mbar"
482             color: "#4b5563"
483             font.family: "Segoe UI"
484             font.pixelSize: 10
485         }
486     }
487 }
488
489 Rectangle {
490     width: 115
491     height: 90
```

```
492         radius: 10
493         color: "#0a0e17"
494         border.color: "#1e2a3f"
495         border.width: 1
496         clip: true
497
498         Rectangle { width: 4; height: parent.height; color: "#9333ea" }
499
500         Column {
501             anchors.top: parent.top
502             anchors.right: parent.right
503             anchors.bottom: parent.bottom
504             anchors.left: parent.left
505             anchors.topMargin: 10
506             anchors.rightMargin: 10
507             anchors.bottomMargin: 10
508             anchors.leftMargin: 16
509             spacing: 6
510
511             Text {
512                 text: "KONUM"
513                 color: "#6b7280"
514                 font.family: "Segoe UI"
515                 font.pixelSize: 10
516                 font.letterSpacing: 1
517             }
518
519             Text {
520                 text: sensorManager.veriGecerli ? sensorManager.konum.toFixed(1) : "--"
521                 color: "#9333ea"
522                 font.family: "Segoe UI"
523                 font.pixelSize: 22
524                 font.bold: true
```

```
525         }
526
527         Text {
528             text: "mm"
529             color: "#4b5563"
530             font.family: "Segoe UI"
531             font.pixelSize: 10
532         }
533     }
534 }
535
536 Rectangle {
537     width: 115
538     height: 90
539     radius: 10
540     color: "#0a0e17"
541     border.color: "#1e2a3f"
542     border.width: 1
543     clip: true
544
545     Rectangle { width: 4; height: parent.height; color: "#f59e0b" }
546
547     Column {
548         anchors.top: parent.top
549         anchors.right: parent.right
550         anchors.bottom: parent.bottom
551         anchors.left: parent.left
552         anchors.topMargin: 10
553         anchors.rightMargin: 10
554         anchors.bottomMargin: 10
555         anchors.leftMargin: 16
556         spacing: 6
557     }
```

```
558         Text {
559             text: "HIZ"
560             color: "#6b7280"
561             font.family: "Segoe UI"
562             font.pixelSize: 10
563             font.letterSpacing: 1
564         }
565
566         Text {
567             text: sensorManager.veriGecerli ? sensorManager.hiz.toFixed(1) : "--"
568             color: "#f59e0b"
569             font.family: "Segoe UI"
570             font.pixelSize: 22
571             font.bold: true
572         }
573
574         Text {
575             text: "m/s"
576             color: "#4b5563"
577             font.family: "Segoe UI"
578             font.pixelSize: 10
579         }
580     }
581 }
582
583 Rectangle {
584     width: 115
585     height: 90
586     radius: 10
587     color: "#0a0e17"
588     border.color: "#1e2a3f"
589     border.width: 1
590     clip: true
```

```
591
592         Rectangle { width: 4; height: parent.height; color: "#16a34a" }
593
594         Column {
595             anchors.top: parent.top
596             anchors.right: parent.right
597             anchors.bottom: parent.bottom
598             anchors.left: parent.left
599             anchors.topMargin: 10
600             anchors.rightMargin: 10
601             anchors.bottomMargin: 10
602             anchors.leftMargin: 16
603             spacing: 6
604
605             Text {
606                 text: "DEBİ"
607                 color: "#6b7280"
608                 font.family: "Segoe UI"
609                 font.pixelSize: 10
610                 font.letterSpacing: 1
611             }
612
613             Text {
614                 text: sensorManager.veriGecerli ? sensorManager.debi.toFixed(1) : "--"
615                 color: "#16a34a"
616                 font.family: "Segoe UI"
617                 font.pixelSize: 22
618                 font.bold: true
619             }
620
621             Text {
622                 text: "m3/h"
623                 color: "#4b5563"
```

```
624             font.family: "Segoe UI"
625             font.pixelSize: 10
626         }
627     }
628 }
629 }
630 }
631
632 Rectangle {
633     id: panelAyraci
634     width: 3
635     height: parent.height
636     color: "#05070b"
637
638     Rectangle {
639         anchors.right: parent.right
640         anchors.top: parent.top
641         anchors.bottom: parent.bottom
642         width: 1
643         color: "#403b82f6"
644     }
645 }
646
647 Item {
648     width: parent.width - testBilgileriPaneli.width - panelAyraci.width
649     height: parent.height
650
651     Grid {
652         anchors.fill: parent
653         anchors.margins: 16
654         columns: 2
655         spacing: 16
656     }
```

```
657         Rectangle {
658             width: (parent.width - parent.spacing) / 2
659             height: (parent.height - parent.spacing) / 2
660             radius: 10
661             color: "#0f1420"
662             border.color: "#17263d"
663             border.width: 1
664
665             Item {
666                 id: basincBaslik
667                 anchors.top: parent.top
668                 anchors.left: parent.left
669                 anchors.right: parent.right
670                 anchors.margins: 14
671                 height: 20
672
673                 Rectangle {
674                     id: basincNoktasi
675                     width: 8
676                     height: 8
677                     radius: 4
678                     color: "#3b82f6"
679                     anchors.left: parent.left
680                     anchors.verticalCenter: parent.verticalCenter
681                 }
682
683                 Text {
684                     anchors.left: basincNoktasi.right
685                     anchors.leftMargin: 8
686                     anchors.verticalCenter: parent.verticalCenter
687                     text: "Basınç - Zaman"
688                     color: "#9ca3af"
689                     font.family: "Segoe UI"
```

```
690         font.pixelSize: 13
691         font.bold: true
692     }
693
694     Text {
695         anchors.right: parent.right
696         anchors.verticalCenter: parent.verticalCenter
697         text: sensorManager.veriGecerli ? sensorManager.basinc.toFixed(1) + " mbar"
698         : "-- mbar"
699
700         color: "#3b82f6"
701         font.family: "Segoe UI"
702         font.pixelSize: 13
703         font.bold: true
704     }
705 }
706
707 ChartView {
708     anchors.top: basincBaslik.bottom
709     anchors.left: parent.left
710     anchors.right: parent.right
711     anchors.bottom: parent.bottom
712     anchors.topMargin: 4
713     anchors.bottomMargin: 4
714     backgroundColor: "transparent"
715     legend.visible: false
716     antialiasing: true
717
718     ValueAxis {
719         id: xEkseni
720         min: 0
721         max: 60
722         gridLineColor: "#182131"
723         labelsColor: "#4b5563"
```



```
722         labelsFont.pixelSize: 9
723         lineVisible: false
724         minorGridVisible: false
725     }
726
727     ValueAxis {
728         id: yEkseni
729         min: 0
730         max: 300
731         gridLineColor: "#182131"
732         labelsColor: "#4b5563"
733         labelsFont.pixelSize: 9
734         lineVisible: false
735         minorGridVisible: false
736     }
737
738     LineSeries {
739         id: basincSerisi
740         axisX: xEkseni
741         axisY: yEkseni
742         color: "#3b82f6"
743         width: 2
744     }
745 }
746
747
748 Rectangle {
749     width: (parent.width - parent.spacing) / 2
750     height: (parent.height - parent.spacing) / 2
751     radius: 10
752     color: "#0f1420"
753     border.color: "#241a38"
754     border.width: 1
```

```
755
756         Item {
757             id: konumBaslik
758             anchors.top: parent.top
759             anchors.left: parent.left
760             anchors.right: parent.right
761             anchors.margins: 14
762             height: 20
763
764             Rectangle {
765                 id: konumNoktasi
766                 width: 8
767                 height: 8
768                 radius: 4
769                 color: "#9333ea"
770                 anchors.left: parent.left
771                 anchors.verticalCenter: parent.verticalCenter
772             }
773
774             Text {
775                 anchors.left: konumNoktasi.right
776                 anchors.leftMargin: 8
777                 anchors.verticalCenter: parent.verticalCenter
778                 text: "Konum - Zaman"
779                 color: "#9ca3af"
780                 font.family: "Segoe UI"
781                 font.pixelSize: 13
782                 font.bold: true
783             }
784
785             Text {
786                 anchors.right: parent.right
787                 anchors.verticalCenter: parent.verticalCenter
```

```
788         text: sensorManager.veriGecerli ? sensorManager.konum.toFixed(1) + " mm" :  
    "-- mm"  
  
789         color: "#9333ea"  
  
790         font.family: "Segoe UI"  
  
791         font.pixelSize: 13  
  
792         font.bold: true  
  
793     }  
  
794 }  
  
795  
  
796     ChartView {  
  
797         anchors.top: konumBaslik.bottom  
  
798         anchors.left: parent.left  
  
799         anchors.right: parent.right  
  
800         anchors.bottom: parent.bottom  
  
801         anchors.topMargin: 4  
  
802         anchors.bottomMargin: 4  
  
803         backgroundColor: "transparent"  
  
804         legend.visible: false  
  
805         antialiasing: true  
  
806  
  
807         ValueAxis {  
  
808             id: konumXEkseni  
  
809             min: 0  
  
810             max: 60  
  
811             gridLineColor: "#182131"  
  
812             labelsColor: "#4b5563"  
  
813             labelsFont.pixelSize: 9  
  
814             lineVisible: false  
  
815             minorGridVisible: false  
  
816         }  
  
817  
  
818         ValueAxis {  
  
819             id: konumYEkseni  
  
820             min: 0
```

```

821                max: 500
822                gridLineColor: "#182131"
823                labelsColor: "#4b5563"
824                labelsFont.pixelSize: 9
825                lineVisible: false
826                minorGridVisible: false
827            }
828
829            LineSeries {
830                id: konumSerisi
831                axisX: konumXEkseni
832                axisY: konumYEkseni
833                color: "#9333ea"
834                width: 2
835            }
836
837            Connections {
838                target: calculator
839                function onStrokeSayisiChanged() {
840                    if (aktifOlcumId > 0 && sensorManager.veriGecerli) {
841                        database.strokeKaydet(aktifOlcumId, sensorManager.basinc,
842                        sensorManager.konum, sensorManager.debi, true)
843                    }
844                }
845            }
846        }
847
848        Rectangle {
849            width: (parent.width - parent.spacing) / 2
850            height: (parent.height - parent.spacing) / 2
851            radius: 10
852            color: "#0f1420"

```

```
853         border.color: "#3a2a14"
854         border.width: 1
855
856         Item {
857             id: hizBaslik
858             anchors.top: parent.top
859             anchors.left: parent.left
860             anchors.right: parent.right
861             anchors.margins: 14
862             height: 20
863
864             Rectangle {
865                 id: hizNoktasi
866                 width: 8
867                 height: 8
868                 radius: 4
869                 color: "#f59e0b"
870                 anchors.left: parent.left
871                 anchors.verticalCenter: parent.verticalCenter
872             }
873
874             Text {
875                 anchors.left: hizNoktasi.right
876                 anchors.leftMargin: 8
877                 anchors.verticalCenter: parent.verticalCenter
878                 text: "Hız - Zaman"
879                 color: "#9ca3af"
880                 font.family: "Segoe UI"
881                 font.pixelSize: 13
882                 font.bold: true
883             }
884
885             Text {
```

```

886                anchors.right: parent.right
887                anchors.verticalCenter: parent.verticalCenter
888                text: sensorManager.veriGecerli ? sensorManager.hiz.toFixed(1) + " m/s" : "-
- m/s"
889                color: "#f59e0b"
890                font.family: "Segoe UI"
891                font.pixelSize: 13
892                font.bold: true
893            }
894        }
895
896        ChartView {
897            anchors.top: hizBaslik.bottom
898            anchors.left: parent.left
899            anchors.right: parent.right
900            anchors.bottom: parent.bottom
901            anchors.topMargin: 4
902            anchors.bottomMargin: 4
903            backgroundColor: "transparent"
904            legend.visible: false
905            antialiasing: true
906
907            ValueAxis {
908                id: hizXEkseni
909                min: 0
910                max: 60
911                gridLineColor: "#182131"
912                labelsColor: "#4b5563"
913                labelsFont.pixelSize: 9
914                lineVisible: false
915                minorGridVisible: false
916            }
917
918            ValueAxis {

```

```
919             id: hizYEkseni
920             min: 0
921             max: 4
922             gridLineColor: "#182131"
923             labelsColor: "#4b5563"
924             labelsFont.pixelSize: 9
925             lineVisible: false
926             minorGridVisible: false
927         }
928
929         LineSeries {
930             id: hizSerisi
931             axisX: hizXEkseni
932             axisY: hizYEkseni
933             color: "#f59e0b"
934             width: 2
935         }
936     }
937 }
938
939 Rectangle {
940     width: (parent.width - parent.spacing) / 2
941     height: (parent.height - parent.spacing) / 2
942     radius: 10
943     color: "#0f1420"
944     border.color: "#163321"
945     border.width: 1
946
947     Item {
948         id: debiBaslik
949         anchors.top: parent.top
950         anchors.left: parent.left
951         anchors.right: parent.right
```

```

952         anchors.margins: 14
953         height: 20
954
955         Rectangle {
956             id: debiNoktasi
957             width: 8
958             height: 8
959             radius: 4
960             color: "#16a34a"
961             anchors.left: parent.left
962             anchors.verticalCenter: parent.verticalCenter
963         }
964
965         Text {
966             anchors.left: debiNoktasi.right
967             anchors.leftMargin: 8
968             anchors.verticalCenter: parent.verticalCenter
969             text: "Debi - Zaman"
970             color: "#9ca3af"
971             font.family: "Segoe UI"
972             font.pixelSize: 13
973             font.bold: true
974         }
975
976         Text {
977             anchors.right: parent.right
978             anchors.verticalCenter: parent.verticalCenter
979             text: sensorManager.veriGecerli ? sensorManager.debi.toFixed(1) + " m3/h" :
    "-- m3/h"
980             color: "#16a34a"
981             font.family: "Segoe UI"
982             font.pixelSize: 13
983             font.bold: true

```



```
984         }
985     }
986
987     ChartView {
988         anchors.top: debiBaslik.bottom
989         anchors.left: parent.left
990         anchors.right: parent.right
991         anchors.bottom: parent.bottom
992         anchors.topMargin: 4
993         anchors.bottomMargin: 4
994         backgroundColor: "transparent"
995         legend.visible: false
996         antialiasing: true
997
998         ValueAxis {
999             id: debiXEkseni
1000             min: 0
1001             max: 60
1002             gridLineColor: "#182131"
1003             labelsColor: "#4b5563"
1004             labelsFont.pixelSize: 9
1005             lineVisible: false
1006             minorGridVisible: false
1007         }
1008
1009         ValueAxis {
1010             id: debiYEkseni
1011             min: 0
1012             max: 180
1013             gridLineColor: "#182131"
1014             labelsColor: "#4b5563"
1015             labelsFont.pixelSize: 9
1016             lineVisible: false
```

```
1017             minorGridVisible: false
1018         }
1019
1020         LineSeries {
1021             id: debiSerisi
1022             axisX: debiXEkseni
1023             axisY: debiYEkseni
1024             color: "#16a34a"
1025             width: 2
1026         }
1027     }
1028 }
1029 }
1030 }
1031 }
1032
1033 Popup {
1034     id: bitirOnayPopup
1035     modal: true
1036     focus: true
1037     anchors.centerIn: parent
1038     width: 340
1039     padding: 24
1040     closePolicy: Popup.CloseOnEscape | Popup.CloseOnPressOutside
1041
1042     background: Rectangle {
1043         color: "#0f1420"
1044         radius: 12
1045         border.color: "#1e2a3f"
1046         border.width: 1
1047     }
1048
1049     Overlay.modal: Rectangle {
```

```
1050         color: "#a6000000"
1051     }
1052
1053     contentItem: Column {
1054         spacing: 20
1055
1056         Text {
1057             width: 292
1058             text: "Ölçümü bitirmek istediğinize emin misiniz?"
1059             color: "#dce8f5"
1060             font.family: "Segoe UI"
1061             font.pixelSize: 14
1062             font.bold: true
1063             wrapMode: Text.WordWrap
1064         }
1065
1066         Row {
1067             width: 292
1068             spacing: 10
1069
1070             Button {
1071                 id: vazgecButonu
1072                 width: (parent.width - parent.spacing) / 2
1073                 height: 40
1074                 text: "Vazgeç"
1075                 font.family: "Segoe UI"
1076                 font.pixelSize: 13
1077
1078                 onClicked: bitirOnayPopup.close()
1079
1080                 background: Rectangle {
1081                     radius: 8
1082                     color: vazgecButonu.pressed ? "#1e2a3f" : (vazgecButonu.hovered ? "#243349" :
"#182131")
```

```
1083         border.color: "#2c3b52"
1084         border.width: 1
1085     }
1086
1087     contentItem: Text {
1088         text: vazgecButonu.text
1089         color: "#dce8f5"
1090         font: vazgecButonu.font
1091         horizontalAlignment: Text.AlignHCenter
1092         verticalAlignment: Text.AlignVCenter
1093     }
1094 }
1095
1096 Button {
1097     id: evetBitirButonu
1098     width: (parent.width - parent.spacing) / 2
1099     height: 40
1100     text: "Evet, Bitir"
1101     font.family: "Segoe UI"
1102     font.pixelSize: 13
1103
1104     onClicked: {
1105         bitirOnayPopup.close()
1106         var bitenId = aktifOlcumId
1107         aktifOlcumId = -1
1108         uyariMesaji = ""
1109         console.log("Olcum sonlandirildi, id:", bitenId)
1110         olcumTamamlandi(bitenId)
1111     }
1112
1113     background: Rectangle {
1114         radius: 8
```

```
1115         color: evetBitirButonu.pressed ? "#7f1d1d" : (evetBitirButonu.hovered ?
    "#b91c1c" : "#991b1b")
1116         border.color: "#dc2626"
1117         border.width: 1
1118     }
1119
1120     contentItem: Text {
1121         text: evetBitirButonu.text
1122         color: "#ffffff"
1123         font: evetBitirButonu.font
1124         horizontalAlignment: Text.AlignHCenter
1125         verticalAlignment: Text.AlignVCenter
1126     }
1127 }
1128 }
1129 }
1130 }
1131 }
```

```
1 import QtQuick 6.7
2 import QtQuick.Controls 6.7
3 import QtCharts 6.7
4
5 Rectangle {
6     color: "#0a0e17"
7
8     property int olcumId: -1
9     property real hesaplananTau0: 0
10    property real hesaplananMu: 0
11    property int playbackAdimi: -1
12    property bool oynatiliyor: false
13    property int playbackHizi: 1
14    property string musteriAdi: ""
15    property string receteAdi: ""
16
17    onOlcumIdChanged: verileriYukle()
18
19    function verileriYukle() {
20        if (olcumId <= 0) {
21            return
22        }
23
24        playbackAdimi = -1
25        oynatiliyor = false
26        playbackTimer.stop()
27
28        var bilgi = database.olcumBilgisiGetir(olcumId)
29        musteriAdi = bilgi.bulundu ? bilgi.musteri : "Bilinmiyor"
30        receteAdi = bilgi.bulundu ? bilgi.recete : "Bilinmiyor"
31    }
```

```
32     var sonuc = database.binghamHesapla(olcumId)
33
34     hesaplananTau0 = sonuc.tau0
35     hesaplananMu = sonuc.mu
36
37     tau0Metni.text = sonuc.tau0.toFixed(2) + " mbar"
38     muMetni.text = sonuc.mu.toFixed(2) + " mbar·h/m³"
39     r2Metni.text = sonuc.r2.toFixed(2)
40
41     durumKutusu.durumIyiMi = sonuc.yeterliVeri && sonuc.tau0 < 5 && sonuc.mu < 10
42     durumKutusu.tau0Yuksek = sonuc.tau0 >= 5
43     durumKutusu.muYuksek = sonuc.mu >= 10
44
45     pqSerisi.clear()
46     regresyonCizgisi.clear()
47     oynatmaSerisi.clear()
48     strokeModeli.clear()
49
50     var strokeListesi = database.strokeVerileriGetir(olcumId)
51     for (var i = 0; i < strokeListesi.length; i++) {
52         var s = strokeListesi[i]
53         pqSerisi.append(s.debi, s.basinc)
54         strokeModeli.append({ stroke: s.stroke, p: s.basinc, q: s.debi, gecerli: s.gecerli })
55     }
56
57     regresyonCizgisi.append(0, sonuc.tau0)
58     regresyonCizgisi.append(20, sonuc.tau0 + sonuc.mu * 20)
59 }
60
61 Timer {
62     id: playbackTimer
63     interval: 1000 / playbackHizi
64     repeat: true
```

```
65     running: false
66
67     onTriggered: {
68         if (playbackAdimi < strokeModeli.count - 1) {
69             playbackAdimi++
70             var satir = strokeModeli.get(playbackAdimi)
71             oynatmaSerisi.clear()
72             oynatmaSerisi.append(satir.q, satir.p)
73         } else {
74             oynatiliyor = false
75             playbackTimer.stop()
76         }
77     }
78 }
79
80 Row {
81     anchors.fill: parent
82     spacing: 0
83
84     Rectangle {
85         id: ozetPaneli
86         width: 280
87         height: parent.height
88         color: "#0f1420"
89
90         Text {
91             anchors.top: parent.top
92             anchors.left: parent.left
93             anchors.margins: 20
94             text: "SONUÇ ÖZETİ"
95             color: "#6b7280"
96             font.family: "Segoe UI"
97             font.pixelSize: 12
```



```
98         font.bold: true
99         font.letterSpacing: 1
100     }
101
102     Flickable {
103         anchors.top: parent.top
104         anchors.left: parent.left
105         anchors.right: parent.right
106         anchors.bottom: parent.bottom
107         anchors.margins: 20
108         anchors.topMargin: 60
109         contentWidth: width
110         contentHeight: sonucKartlari.height
111         clip: true
112
113         Column {
114             id: sonucKartlari
115             width: parent.width
116             spacing: 10
117
118             Rectangle {
119                 width: parent.width
120                 height: 70
121                 radius: 10
122                 color: "#0a0e17"
123                 border.color: "#17263d"
124                 border.width: 1
125                 clip: true
126
127                 Rectangle { width: 4; height: parent.height; color: "#3b82f6" }
128
129                 Column {
130                     anchors.left: parent.left
```

```
131         anchors.verticalCenter: parent.verticalCenter
132         anchors.leftMargin: 20
133         spacing: 2
134
135         Text {
136             text: "AKMA GERİLMESİ ( $\tau_o$ )"
137             color: "#6b7280"
138             font.family: "Segoe UI"
139             font.pixelSize: 10
140             font.letterSpacing: 1
141         }
142
143         Text {
144             id: tau0Metni
145             text: "—"
146             color: "#3b82f6"
147             font.family: "Segoe UI"
148             font.pixelSize: 20
149             font.bold: true
150         }
151     }
152 }
153
154 Rectangle {
155     width: parent.width
156     height: 70
157     radius: 10
158     color: "#0a0e17"
159     border.color: "#241a38"
160     border.width: 1
161     clip: true
162
163     Rectangle { width: 4; height: parent.height; color: "#9333ea" }
```

```
164
165         Column {
166             anchors.left: parent.left
167             anchors.verticalCenter: parent.verticalCenter
168             anchors.leftMargin: 20
169             spacing: 2
170
171             Text {
172                 text: "PLASTİK VISKOZİTE (μ)"
173                 color: "#6b7280"
174                 font.family: "Segoe UI"
175                 font.pixelSize: 10
176                 font.letterSpacing: 1
177             }
178
179             Text {
180                 id: muMetni
181                 text: "-"
182                 color: "#9333ea"
183                 font.family: "Segoe UI"
184                 font.pixelSize: 20
185                 font.bold: true
186             }
187         }
188     }
189
190     Rectangle {
191         width: parent.width
192         height: 70
193         radius: 10
194         color: "#0a0e17"
195         border.color: "#163321"
196         border.width: 1
```

```
197         clip: true
198
199         Rectangle { width: 4; height: parent.height; color: "#16a34a" }
200
201         Column {
202             anchors.left: parent.left
203             anchors.verticalCenter: parent.verticalCenter
204             anchors.leftMargin: 20
205             spacing: 2
206
207             Text {
208                 text: "UYUM KALİTESİ (R²)"
209                 color: "#6b7280"
210                 font.family: "Segoe UI"
211                 font.pixelSize: 10
212                 font.letterSpacing: 1
213             }
214
215             Text {
216                 id: r2Metni
217                 text: "-"
218                 color: "#16a34a"
219                 font.family: "Segoe UI"
220                 font.pixelSize: 20
221                 font.bold: true
222             }
223         }
224     }
225
226     Rectangle {
227         width: parent.width
228         radius: 10
229         color: "#0a0e17"
```

```
230         border.color: "#1e2a3f"
231
232         border.width: 1
233
234         height: tahminIcerik.implicitHeight + 24
235
236     Column {
237
238         id: tahminIcerik
239
240         anchors.top: parent.top
241
242         anchors.left: parent.left
243
244         anchors.right: parent.right
245
246         anchors.margins: 12
247
248         spacing: 8
249
250     Text {
251
252         text: "BORU HATTI TAHMİNİ"
253
254         color: "#6b7280"
255
256         font.family: "Segoe UI"
257
258         font.pixelSize: 10
259
260         font.letterSpacing: 1
261     }
262
263     TextField {
264
265         id: uzunlukKutusu
266
267         width: parent.width
268
269         height: 34
270
271         placeholderText: "Boru Uzunluğu (m)"
272
273         placeholderTextColor: "#4b5563"
274
275         color: "ffffff"
276
277         font.pixelSize: 12
278
279         leftPadding: 10
280
281         verticalAlignment: TextInput.AlignVCenter
282
283         validator: DoubleValidator { bottom: 0; top: 2000; decimals: 0 }
284
285         background: Rectangle {
286
287             color: "#060d17"
```

```

263             radius: 6
264             border.color: uzunlukKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
265             border.width: 1
266         }
267     }
268
269     TextField {
270         id: debiKutusu
271         width: parent.width
272         height: 34
273         placeholderText: "Hedef Debi (m³/h)"
274         placeholderTextColor: "#4b5563"
275         color: "#ffffff"
276         font.pixelSize: 12
277         leftPadding: 10
278         verticalAlignment: TextInput.AlignVCenter
279         validator: DoubleValidator { bottom: 0; top: 200; decimals: 1 }
280         background: Rectangle {
281             color: "#060d17"
282             radius: 6
283             border.color: debiKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
284             border.width: 1
285         }
286     }
287
288     Button {
289         width: parent.width
290         height: 36
291         text: "Tahmini Hesapla"
292         font.pixelSize: 12
293         font.bold: true
294
295         background: Rectangle {

```

```
296         radius: 6
297         color: parent.hovered ? "#4f8cf7" : "#3b82f6"
298         Behavior on color { ColorAnimation { duration: 120 } }
299     }
300
301     contentItem: Text {
302         text: parent.text
303         color: "#ffffff"
304         font: parent.font
305         horizontalAlignment: Text.AlignHCenter
306         verticalAlignment: Text.AlignVCenter
307     }
308
309     onClicked: {
310         var L = parseFloat(uzunlukKutusu.text)
311         var Q = parseFloat(debiKutusu.text)
312
313         if (isNaN(L) || isNaN(Q) || L <= 0) {
314             tahminSonucu.text = "Geçerli değer girin"
315             return
316         }
317
318         var basincMbar = (hesaplananTau0 + hesaplananMu * Q) * (L / 50)
319         var basincBar = basincMbar / 1000
320         tahminSonucu.text = basincBar.toFixed(2) + " bar"
321     }
322 }
323
324 Text {
325     id: tahminSonucu
326     width: parent.width
327     text: "-"
328     color: "#e8a020"
```

```
329         font.family: "Segoe UI"
330         font.pixelSize: 18
331         font.bold: true
332         horizontalAlignment: Text.AlignHCenter
333     }
334 }
335 }
336
337 Rectangle {
338     id: durumKutusu
339     width: parent.width
340     height: durumIcerik.implicitHeight + 24
341     radius: 10
342     color: durumIyiMi ? "#0f2417" : "#2a1414"
343     border.color: durumIyiMi ? "#16a34a" : "#dc2626"
344     border.width: 1
345
346     property bool durumIyiMi: true
347     property bool tau0Yuksek: false
348     property bool muYuksek: false
349
350     Column {
351         id: durumIcerik
352         anchors.top: parent.top
353         anchors.horizontalCenter: parent.horizontalCenter
354         anchors.topMargin: 12
355         spacing: 8
356         width: parent.width - 24
357
358         Row {
359             anchors.horizontalCenter: parent.horizontalCenter
360             spacing: 10
361
```



```

362         Rectangle {
363             width: 24
364             height: 24
365             radius: 12
366             anchors.verticalCenter: parent.verticalCenter
367             color: durumKutusu.durumIyiMi ? "#16a34a" : "#dc2626"
368
369             Text {
370                 anchors.centerIn: parent
371                 text: durumKutusu.durumIyiMi ? "✓" : "X"
372                 color: "#ffffff"
373                 font.pixelSize: 13
374                 font.bold: true
375             }
376         }
377
378         Text {
379             anchors.verticalCenter: parent.verticalCenter
380             text: durumKutusu.durumIyiMi ? "POMPALANABİLİR" : "POMPALAMA SORUNU"
381             color: durumKutusu.durumIyiMi ? "#4ade80" : "#f87171"
382             font.family: "Segoe UI"
383             font.pixelSize: 15
384             font.bold: true
385         }
386     }
387
388     Text {
389         width: parent.width
390         horizontalAlignment: Text.AlignHCenter
391         visible: !durumKutusu.durumIyiMi
392         text: {
393             if (durumKutusu.tau0Yuksek && durumKutusu.muYuksek) {
394                 return "Hem akma gerilmesi hem plastik viskozite yüksek. Su/
çimento oranını artırın ve agrega gradasyonunu gözden geçirin."

```

```
395             } else if (durumKutusu.tau0Yuksek) {
396                 return "Akma gerilmesi yüksek. Su/çimento oranını artırmayı veya
süperakışkanlaştırıcı dozajını yükseltmeyi değerlendirin."
397             } else if (durumKutusu.muYuksek) {
398                 return "Plastik viskozite yüksek. İnce agrega oranını azaltmayı
veya su azaltıcı katkı eklemeyi değerlendirin."
399             }
400             return "Yeterli stroke verisi bulunamadı."
401         }
402         color: "#9ca3af"
403         font.family: "Segoe UI"
404         font.pixelSize: 11
405         wrapMode: Text.WordWrap
406     }
407 }
408 }
409
410 Button {
411     width: parent.width
412     height: 40
413     text: " 📄 PDF Rapor Oluştur"
414     font.pixelSize: 13
415     font.bold: true
416     enabled: olcumId > 0
417
418     onClicked: {
419         reportManager.pdfOlustur(
420             olcumId,
421             musteriAdi,
422             receteAdi,
423             hesaplananTau0,
424             hesaplananMu,
425             parseFloat(r2Metni.text)
426         )
```

```
427         console.log("PDF rapor olusturuldu.")
428     }
429
430     background: Rectangle {
431         radius: 8
432         color: parent.enabled ? (parent.hovered ? "#4f8cf7" : "#3b82f6") : "#1e2a3f"
433         Behavior on color { ColorAnimation { duration: 120 } }
434     }
435
436     contentItem: Text {
437         text: parent.text
438         color: "#ffffff"
439         font: parent.font
440         horizontalAlignment: Text.AlignHCenter
441         verticalAlignment: Text.AlignVCenter
442     }
443 }
444
445 Button {
446     width: parent.width
447     height: 40
448     text: "📄 Excel (CSV) Olarak Dışa Aktar"
449     font.pixelSize: 13
450     font.bold: true
451     enabled: olcumId > 0
452
453     onClicked: {
454         database.csvDisaAktar(olcumId)
455         console.log("CSV disa aktarildi.")
456     }
457
458     background: Rectangle {
459         radius: 8
```

```
460         color: parent.enabled ? (parent.hovered ? "#22c55e" : "#16a34a") : "#1e2a3f"
461         Behavior on color { ColorAnimation { duration: 120 } }
462     }
463
464     contentItem: Text {
465         text: parent.text
466         color: "#ffffff"
467         font: parent.font
468         horizontalAlignment: Text.AlignHCenter
469         verticalAlignment: Text.AlignVCenter
470     }
471 }
472
473 Rectangle {
474     width: parent.width
475     radius: 10
476     color: "#0a0e17"
477     border.color: "#1e2a3f"
478     border.width: 1
479     height: playbackIcerik.implicitHeight + 24
480     visible: olcumId > 0
481
482     Column {
483         id: playbackIcerik
484         anchors.top: parent.top
485         anchors.left: parent.left
486         anchors.right: parent.right
487         anchors.margins: 12
488         spacing: 8
489
490         Text {
491             text: "OYNATMA (PLAYBACK)"
492             color: "#6b7280"
```

```
493         font.family: "Segoe UI"
494         font.pixelSize: 10
495         font.letterSpacing: 1
496     }
497
498     Text {
499         text: strokeModeli.count > 0
500             ? "Stroke " + (playbackAdimi + 1) + " / " + strokeModeli.count
501             : "Oynatılacak stroke yok"
502         color: "#dce8f5"
503         font.family: "Segoe UI"
504         font.pixelSize: 12
505     }
506
507     Row {
508         width: parent.width
509         spacing: 8
510
511         Button {
512             width: 44
513             height: 34
514             text: oynatiliyor ? "||" : "▶"
515             font.pixelSize: 14
516             enabled: strokeModeli.count > 0
517
518             onClicked: {
519                 if (oynatiliyor) {
520                     oynatiliyor = false
521                     playbackTimer.stop()
522                 } else {
523                     if (playbackAdimi >= strokeModeli.count - 1) {
524                         playbackAdimi = -1
525                     }
526                 }
527             }
528         }
529     }
530 }
```

```
526             oynatiliyor = true
527             playbackTimer.start()
528         }
529     }
530
531     background: Rectangle {
532         radius: 6
533         color: parent.enabled ? (parent.hovered ? "#4f8cf7" : "#3b82f6")
534 : "#1e2a3f"
535
536         Behavior on color { ColorAnimation { duration: 120 } }
537     }
538
539     contentItem: Text {
540         text: parent.text
541         color: "#ffffff"
542         font: parent.font
543         horizontalAlignment: Text.AlignHCenter
544         verticalAlignment: Text.AlignVCenter
545     }
546 }
547
548 Repeater {
549     model: [1, 2, 5, 10]
550
551     Button {
552         width: 44
553         height: 34
554         text: modelData + "x"
555         font.pixelSize: 12
556
557         onClicked: playbackHizi = modelData
558
559         background: Rectangle {
```

```

558         radius: 6
559         color: playbackHizi === modelData ? "#3b82f6" : "#0f1420"
560         border.color: "#1e2a3f"
561         border.width: 1
562     }
563
564     contentItem: Text {
565         text: parent.text
566         color: playbackHizi === modelData ? "ffffff" : "#9ca3af"
567         font: parent.font
568         horizontalAlignment: Text.AlignHCenter
569         verticalAlignment: Text.AlignVCenter
570     }
571 }
572 }
573 }
574 }
575 }
576 }
577 }
578 }
579
580 Rectangle {
581     id: panelAyraci
582     width: 3
583     height: parent.height
584     color: "#05070b"
585
586     Rectangle {
587         anchors.right: parent.right
588         anchors.top: parent.top
589         anchors.bottom: parent.bottom
590         width: 1

```

```
591         color: "#403b82f6"
592     }
593 }
594
595 Item {
596     width: parent.width - ozetPaneli.width - panelAyraci.width
597     height: parent.height
598
599     Rectangle {
600         id: grafikKutusu
601         anchors.top: parent.top
602         anchors.left: parent.left
603         anchors.right: parent.right
604         anchors.margins: 16
605         height: parent.height * 0.55
606         radius: 10
607         color: "#0f1420"
608         border.color: "#1e2a3f"
609         border.width: 1
610
611         Item {
612             id: grafikBaslik
613             anchors.top: parent.top
614             anchors.left: parent.left
615             anchors.right: parent.right
616             anchors.margins: 14
617             height: 20
618
619             Text {
620                 anchors.left: parent.left
621                 anchors.verticalCenter: parent.verticalCenter
622                 text: "P-Q Dağılım Grafiği"
623                 color: "#9ca3af"
```



```
624         font.family: "Segoe UI"
625         font.pixelSize: 13
626         font.bold: true
627     }
628
629     Row {
630         anchors.right: parent.right
631         anchors.verticalCenter: parent.verticalCenter
632         spacing: 16
633
634         Row {
635             spacing: 6
636             anchors.verticalCenter: parent.verticalCenter
637
638             Rectangle {
639                 width: 8
640                 height: 8
641                 radius: 4
642                 color: "#3b82f6"
643                 anchors.verticalCenter: parent.verticalCenter
644             }
645
646             Text {
647                 text: "Ölçüm Noktaları"
648                 color: "#6b7280"
649                 font.family: "Segoe UI"
650                 font.pixelSize: 11
651                 anchors.verticalCenter: parent.verticalCenter
652             }
653         }
654
655         Row {
656             spacing: 6
```

```
657         anchors.verticalCenter: parent.verticalCenter
658
659         Rectangle {
660             width: 12
661             height: 2
662             color: "#e8a020"
663             anchors.verticalCenter: parent.verticalCenter
664         }
665
666         Text {
667             text: "Regresyon Doğrusu"
668             color: "#6b7280"
669             font.family: "Segoe UI"
670             font.pixelSize: 11
671             anchors.verticalCenter: parent.verticalCenter
672         }
673     }
674
675     Row {
676         spacing: 6
677         anchors.verticalCenter: parent.verticalCenter
678
679         Rectangle {
680             width: 8
681             height: 8
682             radius: 4
683             color: "#e8a020"
684             anchors.verticalCenter: parent.verticalCenter
685         }
686
687         Text {
688             text: "Oynatma"
689             color: "#6b7280"
```

```

690                font.family: "Segoe UI"
691                font.pixelSize: 11
692                anchors.verticalCenter: parent.verticalCenter
693            }
694        }
695    }
696 }
697
698 ChartView {
699     anchors.top: grafikBaslik.bottom
700     anchors.left: parent.left
701     anchors.right: parent.right
702     anchors.bottom: parent.bottom
703     anchors.topMargin: 4
704     anchors.margins: 8
705     backgroundColor: "transparent"
706     legend.visible: false
707     antialiasing: true
708
709     ValueAxis {
710         id: qEkseni
711         min: 0
712         max: 20
713         titleText: "Debi (Q) m³/h"
714         gridLineColor: "#182131"
715         labelsColor: "#4b5563"
716         labelsFont.pixelSize: 9
717         lineVisible: false
718         minorGridVisible: false
719     }
720
721     ValueAxis {
722         id: pEkseni

```

```
723         min: 0
724         max: 100
725         titleText: "Basınç (P) mbar"
726         gridLineColor: "#182131"
727         labelsColor: "#4b5563"
728         labelsFont.pixelSize: 9
729         lineVisible: false
730         minorGridVisible: false
731     }
732
733     ScatterSeries {
734         id: pqSerisi
735         axisX: qEkseni
736         axisY: pEkseni
737         color: "#3b82f6"
738         markerSize: 10
739     }
740
741     LineSeries {
742         id: regresyonCizgisi
743         axisX: qEkseni
744         axisY: pEkseni
745         color: "#e8a020"
746         width: 2
747     }
748
749     ScatterSeries {
750         id: oynatmaSerisi
751         axisX: qEkseni
752         axisY: pEkseni
753         color: "#e8a020"
754         markerSize: 16
755     }
```

```
756     }
757 }
758
759 Rectangle {
760     anchors.top: grafikKutusu.bottom
761     anchors.left: parent.left
762     anchors.right: parent.right
763     anchors.bottom: parent.bottom
764     anchors.margins: 16
765     anchors.topMargin: 8
766     radius: 10
767     color: "#0f1420"
768     border.color: "#1e2a3f"
769     border.width: 1
770
771     Text {
772         anchors.top: parent.top
773         anchors.left: parent.left
774         anchors.margins: 14
775         text: "Stroke Tablosu"
776         color: "#9ca3af"
777         font.family: "Segoe UI"
778         font.pixelSize: 13
779         font.bold: true
780     }
781
782     Row {
783         id: tabloBasligi
784         anchors.top: parent.top
785         anchors.left: parent.left
786         anchors.right: parent.right
787         anchors.topMargin: 36
788         anchors.leftMargin: 12
```

```

789         anchors.rightMargin: 12
790
791         Text { width: parent.width * 0.25; text: "STROKE"; color: "#6b7280"; font.pixelSize:
11; font.bold: true; font.letterSpacing: 1 }
792
793         Text { width: parent.width * 0.25; text: "P (mbar)"; color: "#6b7280";
font.pixelSize: 11; font.bold: true; font.letterSpacing: 1 }
794
795         Text { width: parent.width * 0.25; text: "Q (m³/h)"; color: "#6b7280";
font.pixelSize: 11; font.bold: true; font.letterSpacing: 1 }
796
797         Text { width: parent.width * 0.25; text: "DURUM"; color: "#6b7280"; font.pixelSize:
11; font.bold: true; font.letterSpacing: 1 }
798
799     }
800
801     Rectangle {
802
803         anchors.top: tabloBasligi.bottom
804
805         anchors.left: parent.left
806
807         anchors.right: parent.right
808
809         anchors.topMargin: 8
810
811         anchors.leftMargin: 12
812
813         anchors.rightMargin: 12
814
815         height: 1
816
817         color: "#1e2a3f"
818
819     }
820
821     ListView {
822
823         anchors.top: tabloBasligi.bottom
824
825         anchors.left: parent.left
826
827         anchors.right: parent.right
828
829         anchors.bottom: parent.bottom
830
831         anchors.margins: 12
832
833         anchors.topMargin: 14
834
835         clip: true
836
837
838         model: ListModel {
839
840             id: strokeModeli
841
842         }

```

```

820
821         delegate: Rectangle {
822             width: parent.width
823             height: 32
824             radius: 6
825             color: index === playbackAdimi ? "#1e2a3f" : (index % 2 === 0 ? "transparent" :
"#0a0e17")
826
827         Row {
828             anchors.fill: parent
829             anchors.leftMargin: 6
830
831             Text { width: parent.width * 0.25; anchors.verticalCenter:
parent.verticalCenter; text: stroke; color: "#dce8f5"; font.pixelSize: 12 }
832             Text { width: parent.width * 0.25; anchors.verticalCenter:
parent.verticalCenter; text: p.toFixed(1); color: "#dce8f5"; font.pixelSize: 12 }
833             Text { width: parent.width * 0.25; anchors.verticalCenter:
parent.verticalCenter; text: q.toFixed(2); color: "#dce8f5"; font.pixelSize: 12 }
834
835         Item {
836             width: parent.width * 0.25
837             height: parent.height
838
839             Rectangle {
840                 anchors.verticalCenter: parent.verticalCenter
841                 width: durumMetni.implicitWidth + 16
842                 height: 20
843                 radius: 10
844                 color: gecerli ? "#123321" : "#2a1414"
845
846                 Text {
847                     id: durumMetni
848                     anchors.centerIn: parent
849                     text: gecerli ? "Geçerli" : "Hatalı"
850                     color: gecerli ? "#4ade80" : "#f87171"

```

```
851             font.pixelSize: 11
852             font.bold: true
853         }
854     }
855 }
856 }
857 }
858 }
859 }
860 }
861 }
862 }
```


src/Backend.h

```
1 #pragma once
2 #include <QObject>
3
4 class Backend : public QObject
5 {
6     Q_OBJECT
7     Q_PROPERTY(QString durum READ durum CONSTANT)
8
9 public:
10     explicit Backend(QObject *parent = nullptr);
11     QString durum() const;
12     Q_INVOKABLE bool girisYap(const QString &kullaniciAdi, const QString &sifre);
13 };
```

src/Backend.cpp

```
1 #include "Backend.h"
2 #include <QDebug>
3
4 Backend::Backend(QObject *parent)
5     : QObject(parent)
6 {
7 }
8
9 QString Backend::durum() const
10 {
11     return "Baglanti bekleniyor...";
12 }
13
14 bool Backend::girisYap(const QString &kullaniciAdi, const QString &sifre)
15 {
16     if (kullaniciAdi == "admin" && sifre == "admin123") {
17         qDebug() << "Giris Basarili:" << kullaniciAdi;
18         return true;
19     }
20
21     qDebug() << "Giris Basarisiz:" << kullaniciAdi;
22     return false;
23 }
```

src/Calculator.h

```
1 #pragma once
2 #include <QObject>
3
4 class Calculator : public QObject
5 {
6     Q_OBJECT
7     Q_PROPERTY(QString durum READ durum NOTIFY durumChanged)
8     Q_PROPERTY(int strokeSayisi READ strokeSayisi NOTIFY strokeSayisiChanged)
9     Q_PROPERTY(bool duraklatildi READ duraklatildi NOTIFY duraklatildiChanged)
10
11 public:
12     explicit Calculator(QObject *parent = nullptr);
13     QString durum() const;
14     int strokeSayisi() const;
15     bool duraklatildi() const;
16
17     Q_INVOKABLE void konumGuncelle(double konum);
18     Q_INVOKABLE void duraklat();
19     Q_INVOKABLE void devamEt();
20     Q_INVOKABLE void sifirla();
21
22 signals:
23     void durumChanged();
24     void strokeSayisiChanged();
25     void duraklatildiChanged();
26
27 private:
28     QString m_durum = "BEKLENIYOR";
29     int m_strokeSayisi = 0;
30     bool m_duraklatildi = false;
31     bool m_yukaridaGorulduMu = false;
```


src/Calculator.cpp

```
1 #include "Calculator.h"
2
3 Calculator::Calculator(QObject *parent)
4     : QObject(parent)
5 {
6 }
7
8 QString Calculator::durum() const
9 {
10     return m_durum;
11 }
12
13 int Calculator::strokeSayisi() const
14 {
15     return m_strokeSayisi;
16 }
17
18 bool Calculator::duraklatildi() const
19 {
20     return m_duraklatildi;
21 }
22
23 void Calculator::konumGuncelle(double konum)
24 {
25     if (m_duraklatildi) {
26         return;
27     }
28
29     QString yeniDurum;
30
31     if (konum > 450) {
```

```
32     yeniDurum = "YUKARIDA";
33 } else if (konum > 20) {
34     yeniDurum = "INIYOR";
35 } else {
36     yeniDurum = "TAMAMLANDI";
37 }
38
39 if (yeniDurum == "YUKARIDA") {
40     m_yukaridaGorulduMu = true;
41 }
42
43 if (yeniDurum == "TAMAMLANDI" && m_durum != "TAMAMLANDI" && m_yukaridaGorulduMu) {
44     m_strokeSayisi++;
45     m_yukaridaGorulduMu = false;
46     emit strokeSayisiChanged();
47 }
48
49 if (yeniDurum != m_durum) {
50     m_durum = yeniDurum;
51     emit durumChanged();
52 }
53 }
54
55 void Calculator::duraklat()
56 {
57     if (m_duraklatildi) {
58         return;
59     }
60
61     m_duraklatildi = true;
62     emit duraklatildiChanged();
63 }
64
```

```
65 void Calculator::devamEt()
66 {
67     if (!m_duraklatildi) {
68         return;
69     }
70
71     m_duraklatildi = false;
72     emit duraklatildiChanged();
73 }
74
75 void Calculator::sifirla()
76 {
77     m_strokeSayisi = 0;
78     m_durum = "BEKLENIYOR";
79     m_duraklatildi = false;
80     m_yukaridaGorulduMu = false;
81     emit strokeSayisiChanged();
82     emit durumChanged();
83     emit duraklatildiChanged();
84 }
```

src/Database.h

```
1 #pragma once
2 #include <QObject>
3 #include <QSqlDatabase>
4 #include <QVariantList>
5 #include <QVariantMap>
6
7 class Database : public QObject
8 {
9     Q_OBJECT
10
11 public:
12     explicit Database(QObject *parent = nullptr);
13
14     Q_INVOKABLE int olcumBaslat(const QString &musteri, const QString &recete, double agirlik);
15     Q_INVOKABLE void strokeKaydet(int olcumId, double basinc, double konum, double debi, bool gecerli);
16     Q_INVOKABLE QVariantList tumOlcumleriGetir();
17     Q_INVOKABLE QVariantList strokeVerileriGetir(int olcumId);
18     Q_INVOKABLE QVariantMap binghamHesapla(int olcumId);
19     Q_INVOKABLE void kalibrasyonKaydet(const QString &sensor, double deger1, double deger2);
20     Q_INVOKABLE QVariantMap kalibrasyonGetir(const QString &sensor);
21     Q_INVOKABLE QVariantMap olcumBilgisiGetir(int olcumId);
22     Q_INVOKABLE bool csvDisaAktar(int olcumId);
23
24 private:
25     void tablolariOlustur();
26     QSqlDatabase m_db;
27 };
```


src/Database.cpp

```
1  #include "Database.h"
2  #include <QSqlQuery>
3  #include <QSqlError>
4  #include <QDebug>
5  #include <QDateTime>
6  #include <QVector>
7  #include <QFile>
8  #include <QTextStream>
9  #include <QStandardPaths>
10 #include <QDir>
11
12 Database::Database(QObject *parent)
13     : QObject(parent)
14 {
15     const QString klasor = QStandardPaths::writableLocation(QStandardPaths::AppLocalDataLocation) +
16         "/database";
17
18     QDir().mkpath(klasor);
19
20     m_db = QSqlDatabase::addDatabase("QSQLITE");
21     m_db.setDatabaseName(klasor + "/sliper.db");
22
23     if (!m_db.open()) {
24         qWarning() << "Veritabani acilamadi:" << m_db.lastError().text();
25     } else {
26         qDebug() << "Veritabani basariyla acildi.";
27         tablolariOlustur();
28     }
29 }
30
31 void Database::tablolariOlustur()
32 {
33     QSqlQuery sorgu;
```

```
32
33     sorgu.exec(
34         "CREATE TABLE IF NOT EXISTS Olcumler ("
35         "id INTEGER PRIMARY KEY AUTOINCREMENT, "
36         "tarih TEXT, "
37         "musteri TEXT, "
38         "recete TEXT, "
39         "agirlik REAL"
40         ")"
41     );
42
43     sorgu.exec(
44         "CREATE TABLE IF NOT EXISTS Strokelar ("
45         "id INTEGER PRIMARY KEY AUTOINCREMENT, "
46         "olcumId INTEGER, "
47         "basinc REAL, "
48         "konum REAL, "
49         "debi REAL, "
50         "gecerli INTEGER, "
51         "FOREIGN KEY(olcumId) REFERENCES Olcumler(id)"
52         ")"
53     );
54
55     QSqlQuery kolonKontrol("PRAGMA table_info(Strokelar)");
56     bool debiKolonuVar = false;
57     while (kolonKontrol.next()) {
58         if (kolonKontrol.value("name").toString() == "debi") {
59             debiKolonuVar = true;
60             break;
61         }
62     }
63
64     if (!debiKolonuVar) {
```

```

65         QSqlQuery migrasyon;
66         migrasyon.exec("ALTER TABLE Stokelar ADD COLUMN debi REAL DEFAULT 0");
67     }
68
69     sorgu.exec(
70         "CREATE TABLE IF NOT EXISTS Kalibrasyonlar ("
71         "sensor TEXT PRIMARY KEY, "
72         "deger1 REAL, "
73         "deger2 REAL, "
74         "tarih TEXT"
75         ")";
76 );
77
78 qDebug() << "Tablolar hazir.";
79 }
80
81 int Database::olcumBaslat(const QString &musteri, const QString &recete, double agirlik)
82 {
83     QSqlQuery sorgu;
84     sorgu.prepare(
85         "INSERT INTO Olcumler (tarih, muster_i, recete, agirlik) "
86         "VALUES (:tarih, :muster_i, :recete, :agirlik)"
87     );
88
89     sorgu.bindValue(":tarih", QDateTime::currentDateTime().toString("dd.MM.yyyy HH:mm"));
90     sorgu.bindValue(":muster_i", muster_i);
91     sorgu.bindValue(":recete", recete);
92     sorgu.bindValue(":agirlik", agirlik);
93
94     if (!sorgu.exec()) {
95         qWarning() << "Olcum baslatilamadi:" << sorgu.lastError().text();
96         return -1;
97     }

```

```
98
99     int yeniId = sorgu.lastInsertId().toInt();
100     qDebug() << "Yeni olcum baslatildi, id:" << yeniId;
101     return yeniId;
102 }
103
104 void Database::strokeKaydet(int olcumId, double basinc, double konum, double debi, bool gecerli)
105 {
106     if (olcumId <= 0) {
107         qWarning() << "Gecersiz olcumId, stroke kaydedilemedi.";
108         return;
109     }
110
111     QSqlQuery sorgu;
112     sorgu.prepare(
113         "INSERT INTO Strokelar (olcumId, basinc, konum, debi, gecerli) "
114         "VALUES (:olcumId, :basinc, :konum, :debi, :gecerli)"
115     );
116
117     sorgu.bindValue(":olcumId", olcumId);
118     sorgu.bindValue(":basinc", basinc);
119     sorgu.bindValue(":konum", konum);
120     sorgu.bindValue(":debi", debi);
121     sorgu.bindValue(":gecerli", gecerli ? 1 : 0);
122
123     if (!sorgu.exec()) {
124         qWarning() << "Stroke kaydedilemedi:" << sorgu.lastError().text();
125     } else {
126         qDebug() << "Stroke kaydedildi, olcumId:" << olcumId;
127     }
128 }
129
130 QVariantList Database::tumOlcumleriGetir()
```

```

131 {
132     QVariantList sonuclar;
133
134     QSqlQuery sorgu(
135         "SELECT o.id, o.tarih, o.musteri, o.recete, o.agirlik, "
136         "COUNT(s.id) AS strokeSayisi "
137         "FROM Olcumlar o "
138         "LEFT JOIN Strokelar s ON s.olcumId = o.id "
139         "GROUP BY o.id "
140         "ORDER BY o.id DESC"
141     );
142
143     while (sorgu.next()) {
144         QVariantMap satir;
145         satir["id"] = sorgu.value("id");
146         satir["tarih"] = sorgu.value("tarih");
147         satir["musteri"] = sorgu.value("musteri");
148         satir["recete"] = sorgu.value("recete");
149         satir["agirlik"] = sorgu.value("agirlik");
150         satir["strokeSayisi"] = sorgu.value("strokeSayisi");
151         sonuclar.append(satir);
152     }
153
154     return sonuclar;
155 }
156
157 QVariantList Database::strokeVerileriGetir(int olcumId)
158 {
159     QVariantList sonuclar;
160
161     QSqlQuery sorgu;
162     sorgu.prepare(
163         "SELECT basinc, konum, debi, gecerli FROM Strokelar "

```

```
164         "WHERE olcumId = :olcumId ORDER BY id ASC"
165     );
166     sorgu.bindValue(":olcumId", olcumId);
167
168     if (!sorgu.exec()) {
169         qWarning() << "Stroke verileri alinamadi:" << sorgu.lastError().text();
170         return sonuclar;
171     }
172
173     int sira = 1;
174     while (sorgu.next()) {
175         QVariantMap satir;
176         satir["stroke"] = sira++;
177         satir["basinc"] = sorgu.value("basinc");
178         satir["konum"] = sorgu.value("konum");
179         satir["debi"] = sorgu.value("debi");
180         satir["gecerli"] = sorgu.value("gecerli").toBool();
181         sonuclar.append(satir);
182     }
183
184     return sonuclar;
185 }
186
187 QVariantMap Database::binghamHesapla(int olcumId)
188 {
189     QVariantMap sonuc;
190
191     QSqlQuery sorgu;
192     sorgu.prepare(
193         "SELECT basinc, debi FROM Strokelar "
194         "WHERE olcumId = :olcumId AND gecerli = 1"
195     );
196     sorgu.bindValue(":olcumId", olcumId);
```

```
197
198     QVector<double> pDegerleri;
199     QVector<double> qDegerleri;
200
201     if (sorgu.exec()) {
202         while (sorgu.next()) {
203             pDegerleri.append(sorgu.value("basinc").toDouble());
204             qDegerleri.append(sorgu.value("debi").toDouble());
205         }
206     }
207
208     const int n = pDegerleri.size();
209
210     if (n < 2) {
211         sonuc["tau0"] = 0.0;
212         sonuc["mu"] = 0.0;
213         sonuc["r2"] = 0.0;
214         sonuc["yeterliVeri"] = false;
215         return sonuc;
216     }
217
218     double qToplam = 0, pToplam = 0, qpToplam = 0, qKareToplam = 0;
219     for (int i = 0; i < n; ++i) {
220         qToplam += qDegerleri[i];
221         pToplam += pDegerleri[i];
222         qpToplam += qDegerleri[i] * pDegerleri[i];
223         qKareToplam += qDegerleri[i] * qDegerleri[i];
224     }
225
226     const double qOrt = qToplam / n;
227     const double pOrt = pToplam / n;
228
229     const double payda = qKareToplam - n * qOrt * qOrt;
```

```

230     const double mu = (payda != 0.0) ? (qpToplam - n * qOrt * pOrt) / payda : 0.0;
231     const double tau0 = pOrt - mu * qOrt;
232
233     double ssTot = 0, ssRes = 0;
234     for (int i = 0; i < n; ++i) {
235         const double tahmin = tau0 + mu * qDegerleri[i];
236         ssRes += (pDegerleri[i] - tahmin) * (pDegerleri[i] - tahmin);
237         ssTot += (pDegerleri[i] - pOrt) * (pDegerleri[i] - pOrt);
238     }
239
240     const double r2 = (ssTot != 0.0) ? (1.0 - ssRes / ssTot) : 0.0;
241
242     sonuc["tau0"] = tau0;
243     sonuc["mu"] = mu;
244     sonuc["r2"] = r2;
245     sonuc["yeterliVeri"] = true;
246     return sonuc;
247 }
248
249 void Database::kalibrasyonKaydet(const QString &sensor, double deger1, double deger2)
250 {
251     QSqlQuery sorgu;
252     sorgu.prepare(
253         "INSERT INTO Kalibrasyonlar (sensor, deger1, deger2, tarih) "
254         "VALUES (:sensor, :deger1, :deger2, :tarih) "
255         "ON CONFLICT(sensor) DO UPDATE SET "
256         "deger1 = :deger1, deger2 = :deger2, tarih = :tarih"
257     );
258
259     sorgu.bindValue(":sensor", sensor);
260     sorgu.bindValue(":deger1", deger1);
261     sorgu.bindValue(":deger2", deger2);
262     sorgu.bindValue(":tarih", QDateTime::currentDateTime().toString("dd.MM.yyyy HH:mm"));

```



```
263
264     if (!sorgu.exec()) {
265         qWarning() << "Kalibrasyon kaydedilemedi:" << sorgu.lastError().text();
266     } else {
267         qDebug() << "Kalibrasyon kaydedildi:" << sensor;
268     }
269 }
270
271 QVariantMap Database::kalibrasyonGetir(const QString &sensor)
272 {
273     QVariantMap sonuc;
274
275     QSqlQuery sorgu;
276     sorgu.prepare("SELECT deger1, deger2, tarih FROM Kalibrasyonlar WHERE sensor = :sensor");
277     sorgu.bindValue(":sensor", sensor);
278
279     if (sorgu.exec() && sorgu.next()) {
280         sonuc["deger1"] = sorgu.value("deger1");
281         sonuc["deger2"] = sorgu.value("deger2");
282         sonuc["tarih"] = sorgu.value("tarih");
283         sonuc["mevcut"] = true;
284     } else {
285         sonuc["mevcut"] = false;
286     }
287
288     return sonuc;
289 }
290
291 QVariantMap Database::olcumBilgisiGetir(int olcumId)
292 {
293     QVariantMap sonuc;
294
295     QSqlQuery sorgu;
```

```
296     sorgu.prepare("SELECT tarih, musteri, recete, agirlik FROM Olcumler WHERE id = :id");
297     sorgu.bindValue(":id", olcumId);
298
299     if (sorgu.exec() && sorgu.next()) {
300         sonuc["tarih"] = sorgu.value("tarih");
301         sonuc["musteri"] = sorgu.value("musteri");
302         sonuc["recete"] = sorgu.value("recete");
303         sonuc["agirlik"] = sorgu.value("agirlik");
304         sonuc["bulundu"] = true;
305     } else {
306         sonuc["bulundu"] = false;
307     }
308
309     return sonuc;
310 }
311
312 bool Database::csvDisaAktar(int olcumId)
313 {
314     QVariantMap bilgi = olcumBilgisiGetir(olcumId);
315     if (!bilgi["bulundu"].toBool()) {
316         qWarning() << "Olcum bulunamadi, csv olusturulamadi.";
317         return false;
318     }
319
320     QString klasor = QStandardPaths::writableLocation(QStandardPaths::DocumentsLocation) +
"/SliperRaporlari";
321     QDir().mkpath(klasor);
322
323     QString dosyaAdi = klasor + QString("/SLIPER_Veri_%1_%2.csv")
324         .arg(olcumId)
325         .arg(QDateTime::currentDateTime().toString("ddMMyyyy_HHmss"));
326
327     QFile dosya(dosyaAdi);
```

```
328     if (!dosya.open(QIODevice::WriteOnly | QIODevice::Text)) {
329         qWarning() << "CSV dosyasi acilamadi:" << dosyaAdi;
330         return false;
331     }
332
333     QTextStream akis(&dosya);
334     akis.setEncoding(QStringConverter::Utf8);
335
336     akis << "Olcum ID;" << olcumId << "\n";
337     akis << "Tarih;" << bilgi["tarih"].toString() << "\n";
338     akis << "Musteri;" << bilgi["musteri"].toString() << "\n";
339     akis << "Recete;" << bilgi["recete"].toString() << "\n";
340     akis << "Agirlik (kg);" << bilgi["agirlik"].toDouble() << "\n";
341     akis << "\n";
342     akis << "Stroke;Basinc (mbar);Konum (mm);Debi (m3/h);Gecerli\n";
343
344     QVariantList strokelar = strokeVerileriGetir(olcumId);
345     for (const QVariant &v : strokelar) {
346         QVariantMap s = v.toMap();
347         akis << s["stroke"].toInt() << ";"
348             << s["basinc"].toDouble() << ";"
349             << s["konum"].toDouble() << ";"
350             << s["debi"].toDouble() << ";"
351             << (s["gecerli"].toBool() ? "Evet" : "Hayir") << "\n";
352     }
353
354     dosya.close();
355     qDebug() << "CSV disa aktarildi:" << dosyaAdi;
356     return true;
357 }
```

src/ReportManager.h

```
1 #pragma once
2 #include <QObject>
3
4 class ReportManager : public QObject
5 {
6     Q_OBJECT
7
8 public:
9     explicit ReportManager(QObject *parent = nullptr);
10    Q_INVOKABLE bool pdfOlustur(int olcumId, const QString &musteri, const QString &recete,
11                                double tau0, double mu, double r2);
12 };
```

src/ReportManager.cpp

```
1 #include "ReportManager.h"
2 #include <QPrinter>
3 #include <QPainter>
4 #include <QTextDocument>
5 #include <QStandardPaths>
6 #include <QDir>
7 #include <QDateTime>
8 #include <QDebug>
9
10 ReportManager::ReportManager(QObject *parent)
11     : QObject(parent)
12 {
13 }
14
15 bool ReportManager::pdfOlustur(int olcumId, const QString &musteri, const QString &recete,
16                                 double tau0, double mu, double r2)
17 {
18     QString klasor = QStandardPaths::writableLocation(QStandardPaths::DocumentsLocation) +
19         "/SliperRaporlari";
20
21     QDir().mkpath(klasor);
22
23     QString dosyaAdi = klasor + QString("/SLIPER_Rapor_%1_%2.pdf")
24         .arg(olcumId)
25         .arg(QDateTime::currentDateTime().toString("ddMMyyyy_HHmmss"));
26
27     QTextDocument belge;
28
29     QString html = QString(
30         "<h1 style='color:#3b82f6;'>SLIPER Analiz Raporu</h1>"
31         "<p><b>Olcum ID:</b> %1</p>"
32         "<p><b>Musteri:</b> %2</p>"
33         "<p><b>Recete:</b> %3</p>"
34         "<p><b>Tarih:</b> %4</p>"
35     );
```

```
32     "<hr>"
33     "<h2>Bingham Model Sonuclari</h2>"
34     "<p><b>Akma Gerilmesi ( $\tau_0$ ):</b> %5 mbar</p>"
35     "<p><b>Plastik Viskozite ( $\mu$ ):</b> %6 mbar*h/m3</p>"
36     "<p><b>Uyum Kalitesi ( $R_2$ ):</b> %7</p>"
37     "<hr>"
38     "<p style='color:#6b7280; font-size:10px;'>Liya Laboratuvar Test Cihazlari - SLIPER Analiz Sistemi</p>"
39     ).arg(olcumId)
40     .arg(musteri)
41     .arg(recete)
42     .arg(QDateTime::currentDateTime().toString("dd.MM.yyyy HH:mm"))
43     .arg(QString::number(tau0, 'f', 2))
44     .arg(QString::number(mu, 'f', 2))
45     .arg(QString::number(r2, 'f', 2));
46
47     belge.setHtml(html);
48
49     QPainter yazici(QPrinter::HighResolution);
50     yazici.setOutputFormat(QPrinter::PdfFormat);
51     yazici.setOutputFileName(dosyaAdi);
52
53     belge.print(&yazici);
54
55     qDebug() << "PDF rapor olusturuldu:" << dosyaAdi;
56     return true;
57 }
```

src/SensorManager.h

```
1 #pragma once
2 #include <QObject>
3
4 class SensorManager : public QObject
5 {
6     Q_OBJECT
7     Q_PROPERTY(double basinc READ basinc NOTIFY basincChanged)
8     Q_PROPERTY(double konum READ konum NOTIFY konumChanged)
9     Q_PROPERTY(double hiz READ hiz NOTIFY hizChanged)
10    Q_PROPERTY(double debi READ debi NOTIFY debiChanged)
11    Q_PROPERTY(double egimX READ egimX NOTIFY egimXChanged)
12    Q_PROPERTY(double egimY READ egimY NOTIFY egimYChanged)
13    Q_PROPERTY(double hamAgirlik READ hamAgirlik NOTIFY hamAgirlikChanged)
14    Q_PROPERTY(bool veriGecerli READ veriGecerli NOTIFY veriGecerliChanged)
15
16 public:
17     explicit SensorManager(QObject *parent = nullptr);
18
19     double basinc() const;
20     double konum() const;
21     double hiz() const;
22     double debi() const;
23     double egimX() const;
24     double egimY() const;
25     double hamAgirlik() const;
26     bool veriGecerli() const;
27
28     Q_INVOKABLE void veriGuncelle(double basinc, double konum, double hiz, double debi, double egimX,
29     double egimY);
30     Q_INVOKABLE void hamAgirlikGuncelle(double hamAgirlik);
31     Q_INVOKABLE void veriyiGecersizYap();
32 }
```

```
32 signals:
33     void basincChanged();
34     void konumChanged();
35     void hizChanged();
36     void debiChanged();
37     void egimXChanged();
38     void egimYChanged();
39     void hamAgirlikChanged();
40     void veriGecerliChanged();
41     void veriGuncellendi();
42
43 private:
44     double m_basinc = 0.0;
45     double m_konum = 0.0;
46     double m_hiz = 0.0;
47     double m_debi = 0.0;
48     double m_egimX = 0.0;
49     double m_egimY = 0.0;
50     double m_hamAgirlik = 0.0;
51     bool m_veriGecerli = false;
52 };
```


src/SensorManager.cpp

```
1 #include "SensorManager.h"
2
3 SensorManager::SensorManager(QObject *parent)
4     : QObject(parent)
5 {
6 }
7
8 double SensorManager::basinc() const { return m_basinc; }
9 double SensorManager::konum() const { return m_konum; }
10 double SensorManager::hiz() const { return m_hiz; }
11 double SensorManager::debi() const { return m_debi; }
12 double SensorManager::egimX() const { return m_egimX; }
13 double SensorManager::egimY() const { return m_egimY; }
14 double SensorManager::hamAgirlik() const { return m_hamAgirlik; }
15 bool SensorManager::veriGecerli() const { return m_veriGecerli; }
16
17 void SensorManager::veriGuncelle(double basinc, double konum, double hiz, double debi, double egimX,
18     double egimY)
19 {
20     const bool veriGecerliDegisti = !m_veriGecerli;
21     m_veriGecerli = true;
22
23     if (m_basinc != basinc) { m_basinc = basinc; emit basincChanged(); }
24     if (m_konum != konum) { m_konum = konum; emit konumChanged(); }
25     if (m_hiz != hiz) { m_hiz = hiz; emit hizChanged(); }
26     if (m_debi != debi) { m_debi = debi; emit debiChanged(); }
27     if (m_egimX != egimX) { m_egimX = egimX; emit egimXChanged(); }
28     if (m_egimY != egimY) { m_egimY = egimY; emit egimYChanged(); }
29     if (veriGecerliDegisti) { emit veriGecerliChanged(); }
30
31     emit veriGuncellendi();
32 }
```

```
32
33 void SensorManager::hamAgirlikGuncelle(double hamAgirlik)
34 {
35     if (m_hamAgirlik != hamAgirlik) {
36         m_hamAgirlik = hamAgirlik;
37         emit hamAgirlikChanged();
38     }
39 }
40
41 void SensorManager::veriyiGecersizYap()
42 {
43     if (!m_veriGecerli) return;
44     m_veriGecerli = false;
45     emit veriGecerliChanged();
46 }
```

src/WifiManager.h

```
1  #pragma once
2  #include <QObject>
3  #include <QTcpSocket>
4  #include <QElapsedTimer>
5  #include "SensorManager.h"
6  #include "Database.h"
7
8  class WifiManager : public QObject
9  {
10     Q_OBJECT
11     Q_PROPERTY(bool baglandi READ baglandi NOTIFY baglandiChanged)
12     Q_PROPERTY(bool baglaniyor READ baglaniyor NOTIFY baglaniyorChanged)
13     Q_PROPERTY(QString durumMesaji READ durumMesaji NOTIFY durumMesajiChanged)
14
15 public:
16     explicit WifiManager(SensorManager *sensorManager, Database *database, QObject *parent = nullptr);
17
18     bool baglandi() const;
19     bool baglaniyor() const;
20     QString durumMesaji() const;
21
22     Q_INVOKABLE void baglan();
23     Q_INVOKABLE void baglantiyiKes();
24
25 signals:
26     void baglandiChanged();
27     void baglaniyorChanged();
28     void durumMesajiChanged();
29
30 private slots:
31     void soketBaglandi();
```

```
32     void soketAyrildi();
33     void soketHata(QAbstractSocket::SocketError hata);
34     void veriHazir();
35
36 private:
37     void durumGuncelle(const QString &mesaj);
38     void jsonSatiriIsle(const QByteArray &satir);
39     void egimHesapla(double accelX, double accelY, double accelZ, double &egimX, double &egimY) const;
40     void kalibrasyonYukle();
41
42     QTcpSocket *m_soket = nullptr;
43     SensorManager *m_sensorManager = nullptr;
44     Database *m_database = nullptr;
45     QByteArray m_tamponVeri;
46
47     bool m_baglandi = false;
48     bool m_baglaniyor = false;
49     QString m_durumMesaji = "Bağlı Değil";
50
51     bool m_ilkPaket = true;
52     double m_oncekiKonum = 0.0;
53     qint64 m_oncekiZamanMs = 0;
54     QElapsedTimer m_zamanlayici;
55
56     // --- Kayitli kalibrasyon degerleri (Database'den yuklenir) ---
57     double m_loadCellKatsayi = 2280.0;
58     double m_loadCellSifirOffset = 0.0;
59     double m_egimOffsetX = 0.0;
60     double m_egimOffsetY = 0.0;
61
62     static constexpr const char *ESP32_IP = "192.168.4.1";
63     static constexpr quint16 ESP32_PORT = 8888;
64
```

```
65     static constexpr double BORU_CAPI_M = 0.126;
```

```
66 };
```

src/WifiManager.cpp

```
1 #include "WifiManager.h"
2 #include <QJsonDocument>
3 #include <QJsonObject>
4 #include <QDebug>
5 #include <cmath>
6
7 WifiManager::WifiManager(SensorManager *sensorManager, Database *database, QObject *parent)
8     : QObject(parent)
9     , m_sensorManager(sensorManager)
10    , m_database(database)
11 {
12     m_soket = new QTcpSocket(this);
13
14     connect(m_soket, &QTcpSocket::connected, this, &WifiManager::soketBaglandi);
15     connect(m_soket, &QTcpSocket::disconnected, this, &WifiManager::soketAyrildi);
16     connect(m_soket, &QTcpSocket::errorOccurred, this, &WifiManager::soketHata);
17     connect(m_soket, &QTcpSocket::readyRead, this, &WifiManager::veriHazir);
18 }
19
20 bool WifiManager::baglandi() const { return m_baglandi; }
21 bool WifiManager::baglaniyor() const { return m_baglaniyor; }
22 QString WifiManager::durumMesaji() const { return m_durumMesaji; }
23
24 void WifiManager::baglan()
25 {
26     if (m_baglandi || m_baglaniyor) return;
27
28     m_baglaniyor = true;
29     emit baglaniyorChanged();
30     durumGuncelle("SLIPER-ESP32'ye baglaniliyor...");
31 }
```

```
32     m_soket->connectToHost(QString(ESP32_IP), ESP32_PORT);
33 }
34
35 void WifiManager::baglantiyiKes()
36 {
37     m_soket->disconnectFromHost();
38 }
39
40 void WifiManager::kalibrasyonYukle()
41 {
42     if (!m_database) return;
43
44     QVariantMap loadCell = m_database->kalibrasyonGetir("loadcell");
45     if (loadCell.value("mevcut").toBool()) {
46         m_loadCellKatsayi = loadCell.value("deger1").toDouble();
47         m_loadCellSifirOffset = loadCell.value("deger2").toDouble();
48         qDebug() << "Load cell kalibrasyonu yuklendi: katsayi=" << m_loadCellKatsayi << "offset=" <<
m_loadCellSifirOffset;
49     }
50
51     QVariantMap egim = m_database->kalibrasyonGetir("egim");
52     if (egim.value("mevcut").toBool()) {
53         m_egimOffsetX = egim.value("deger1").toDouble();
54         m_egimOffsetY = egim.value("deger2").toDouble();
55         qDebug() << "Egim kalibrasyonu yuklendi: X=" << m_egimOffsetX << "Y=" << m_egimOffsetY;
56     }
57 }
58
59 void WifiManager::soketBaglandi()
60 {
61     m_baglandi = true;
62     m_baglaniyor = false;
63     emit baglandiChanged();
64     emit baglaniyorChanged();
```

```
65     durumGuncelle("SLIPER-ESP32 Bağlı");
66
67     m_ilkPaket = true;
68     m_zamanlayici.restart();
69     kalibrasyonYukle();
70
71     if (m_sensorManager) {
72         m_sensorManager->veriyiGecersizYap();
73     }
74
75     qDebug() << "ESP32'ye baglanildi:" << ESP32_IP << ESP32_PORT;
76 }
77
78 void WifiManager::soketAyrildi()
79 {
80     m_baglandi = false;
81     m_baglaniyor = false;
82     emit baglandiChanged();
83     emit baglaniyorChanged();
84     durumGuncelle("Bağlı Değil");
85
86     if (m_sensorManager) {
87         m_sensorManager->veriyiGecersizYap();
88     }
89
90     qDebug() << "ESP32 baglantisi kesildi.";
91 }
92
93 void WifiManager::soketHata(QAbstractSocket::SocketError hata)
94 {
95     m_baglandi = false;
96     m_baglaniyor = false;
97     emit baglandiChanged();
```



```
98     emit baglaniyorChanged();
99     durumGuncelle("Baglanti hatasi: " + m_soket->errorString());
100
101     qWarning() << "Wifi soket hatasi:" << hata << m_soket->errorString();
102 }
103
104 void WifiManager::veriHazir()
105 {
106     m_tamponVeri.append(m_soket->readAll());
107
108     int satirSonu;
109     while ((satirSonu = m_tamponVeri.indexOf('\n')) != -1) {
110         QByteArray satir = m_tamponVeri.left(satirSonu).trimmed();
111         m_tamponVeri.remove(0, satirSonu + 1);
112
113         if (!satir.isEmpty()) {
114             jsonSatiriIsle(satir);
115         }
116     }
117 }
118
119 void WifiManager::egimHesapla(double accelX, double accelY, double accelZ, double &egimX, double &egimY)
120 const
121 {
122     egimX = std::atan2(accelY, accelZ) * 180.0 / M_PI;
123     egimY = std::atan2(-accelX, std::sqrt(accelY * accelY + accelZ * accelZ)) * 180.0 / M_PI;
124 }
125
126 void WifiManager::jsonSatiriIsle(const QByteArray &satir)
127 {
128     QJsonParseError hata;
129     QJsonDocument belge = QJsonDocument::fromJson(satir, &hata);
```

```
130     if (hata.error != QJsonParseError::NoError || !belge.isObject()) {
131         qWarning() << "JSON parse hatasi:" << hata.errorString() << satir;
132         return;
133     }
134
135     QJsonObject obj = belge.object();
136
137     const double hamAgirlik = obj.value("hamAgirlik").toDouble();
138     const double konum = obj.value("konum").toDouble();
139     const double accelX = obj.value("accelX").toDouble();
140     const double accelY = obj.value("accelY").toDouble();
141     const double accelZ = obj.value("accelZ").toDouble();
142
143     // Kalibrasyon ekraninda gerekiyor: ham deger her zaman guncellensin
144     if (m_sensorManager) {
145         m_sensorManager->hamAgirlikGuncelle(hamAgirlik);
146     }
147
148     // --- Basinc: kayitli kalibrasyon ile ---
149     const double agirlikGram = (hamAgirlik - m_loadCellSifirOffset) / m_loadCellKatsayi;
150     const double basinc = agirlikGram * 0.0981;
151
152     // --- Hiz ve debi: konum farkindan turetiliyor ---
153     double hiz = 0.0;
154     double debi = 0.0;
155
156     const qint64 simdikiZamanMs = m_zamanlayici.elapsed();
157
158     if (!m_ilkPaket) {
159         double dt = (simdikiZamanMs - m_oncekiZamanMs) / 1000.0;
160         if (dt <= 0.0) dt = 0.001;
161
162         const double boruAlaniM2 = M_PI * (BORU_CAPI_M / 2.0) * (BORU_CAPI_M / 2.0);
```

```
163     hiz = std::fabs(konum - m_oncekiKonum) / 1000.0 / dt;
164     debi = hiz * boruAlaniM2 * 3600.0;
165 } else {
166     m_ilkPaket = false;
167 }
168
169 m_oncekiKonum = konum;
170 m_oncekiZamanMs = simdikiZamanMs;
171
172 // --- Egim: ham ivmeden aci, sonra kayitli ofset uygulanir ---
173 double egimXHam = 0.0;
174 double egimYHam = 0.0;
175 egimHesapla(accelX, accelY, accelZ, egimXHam, egimYHam);
176
177 const double egimX = egimXHam - m_egimOffsetX;
178 const double egimY = egimYHam - m_egimOffsetY;
179
180 if (m_sensorManager) {
181     m_sensorManager->veriGuncelle(basinc, konum, hiz, debi, egimX, egimY);
182 }
183 }
184
185 void WifiManager::durumGuncelle(const QString &mesaj)
186 {
187     if (m_durumMesaji != mesaj) {
188         m_durumMesaji = mesaj;
189         emit durumMesajiChanged();
190     }
191 }
```